$OneMillion-Bench:
How Far are Language Agents from Human Experts?

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Abstract

As language models (LMs) evolve from chat assistants to long-horizon agents capable of multi-step reasoning and tool use, existing benchmarks remain largely confined to structured or exam-style tasks that fall short of real-world professional demands. To this end, we introduce **$OneMillion-Bench ($1M-Bench)**, a benchmark of 400 expert-curated tasks spanning Law, Finance, Industry, Healthcare, and Natural Science, built to evaluate agents across economically consequential scenarios. Unlike prior work, the benchmark requires retrieving authoritative sources, resolving conflicting evidence, applying domain-specific rules, and making constraint decisions, where correctness depends as much on the reasoning process as the final answer. We adopt a rubric-based evaluation protocol scoring factual accuracy, logical coherence, practical feasibility, and professional compliance, focused on expert-level problems to ensure meaningful differentiation across agents. Together, $1M-Bench provides a unified testbed for assessing agentic reliability, professional depth, and practical readiness in domain-intensive scenarios.

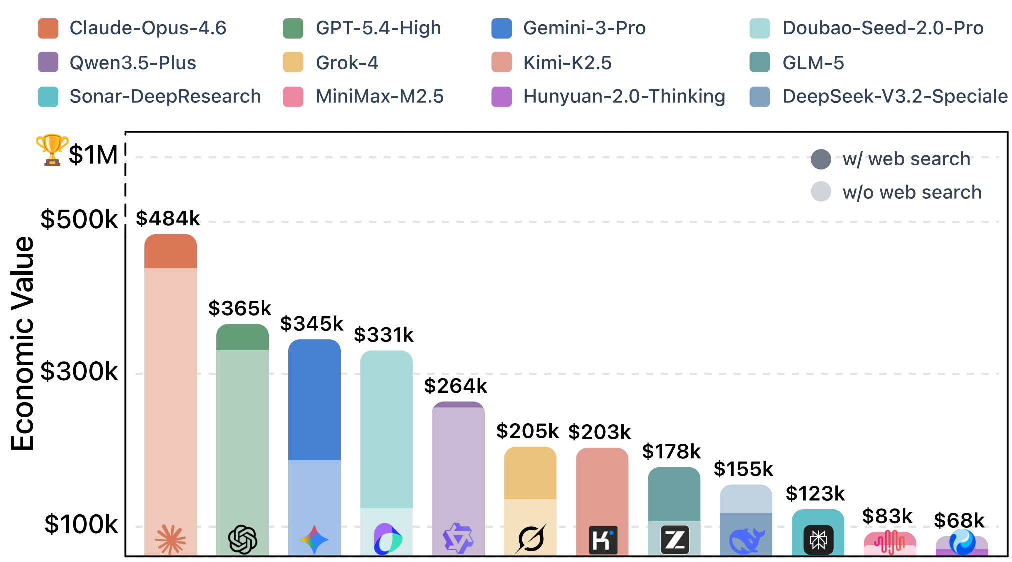

Figure 1: Leaderboard performance on $OneMillion-Bench.

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"If AI learns to gather the scattered sixpences, human gains the liberty to look up at the moon."

1 Introduction

Consider the kinds of work performed daily across industries: a senior actuary auditing a life-insurance reserve valuation under IFRS 17; an M&A lawyer verifying cross-border compliance clauses; or an investment analyst constructing a high-pressure valuation model. These tasks are not merely prompts to be answered, but context-heavy deliverables that demand specialized knowledge and sustained, multi-step reasoning under strict professional constraints. While the agentic capabilities of language models (LMs) have scaled to support multi-step reasoning [6, 7, 12, 17, 32], multi-turn interactions [3, 38, 39], and complex decision-making in controlled settings [2, 42], a critical rift remains between exam-style benchmarks and the rigorous demands of such professional labor. As traditional benchmarks reach saturation [11, 16, 30], it remains fundamentally unclear whether today's agents can reliably create value in economically valued, professional environments.

To effectively quantify the economic value that language agents can deliver in real-world tasks, we introduce **$OneMillion-Bench**, a comprehensive benchmark built around expert-level tasks designed to measure not only accuracy, but value. It spans multiple high-impact professional domains, demands intensive retrieval and evidence-grounded reasoning, and probes agentic reliability under explicit constraints and communication requirements. The benchmark comprises 400 highly challenging open-ended tasks across five high-stakes domains: Finance, Law, Healthcare, Natural Science, and Industry, curated through a collaborative effort spanning over 2,000 expert hours. Each task is assigned a real-world monetary value, calculated by the estimated time required for task completion of a senior professional and its prevailing market hourly wage[1 2] . The estimated value of all tasks exceeds $1 million — giving the benchmark its name.

$OneMillion-Bench offers several key features : 1) **Economic-grounded evaluation** that quantifies agentic capabilities through the lens of tangible labour costs. This clearly distinguish the practical capabilities of LM agents through how much they earn in million-value professional tasks. 2) **Comprehensive domain coverage** spanning five expert professional areas, with balanced coverage and sub-domain partitioning. 3) **Skill-oriented taxonomy** including search, reasoning, verbalization, and instruction following that enables fine-grained analysis beyond aggregate accuracy. This is critical for testing agents across a wide range of real-world settings, rather than overfitting evaluation to any single skill cluster or domain idiosyncrasy. 4) **Rubrics-based evaluation** mechanism to avoid reward hacking and well align with domain-specific policies and expectation designed by domain experts to provide multi-dimensional, fine-grained automated scoring of open agent task performance.

Ultimately, **$OneMillion-Bench** is a step toward value-faithful evaluation of LM agents in professional reality which turns "capability" into an interpretable quantity: how much reliable work an agent can actually deliver, and how much that work is worth, accelerating progress toward AI systems that are *not only powerful, but trustworthy and economically meaningful*.

2 How does $OneMillion-Bench Measure?

2.1 Economic Value

The economic value of $OneMillion-Bench is estimated by multiplying the time senior experts require to resolve each task by their hourly wages:

$$V_{\text{\$OneMillion-Bench}} = T_{\text{ExpertCost}} \times W_{\text{HourlyWage}}$$

[1] https://data.bls.gov/oes/#/industry/000000

[2] https://rsj.gz.gov.cn/zwdt/tzgg/content/post_10619688.html

Expert Cost. To ensure our valuation is grounded in reality, we gather time estimates for each task creation from 2 to 3 senior experts in their respective domains. This collaborative approach minimizes subjective bias, allowing us to derive a reliable mean estimation for task completion times.

Wage Anchoring. We developed a robust and multi-faceted wage reference that considers regional and sectoral nuances. For instance, in the United States, we utilized the latest Occupational Employment and Wage Statistics (OEWS) from the U.S. Bureau of Labor Statistics, complementing it with industry-specific reports for high-income roles, such as financial analysts[3] and medical specialists [4, 15]. In China, we tailored our estimates to reflect the economic realities of tier-1 cities, drawing data from recent labor market wage guidelines.

All calculations are standardized to an hourly basis using a typical work year of 2,080 hours (52 weeks $\times 5$ days $\times 8$ hours). We infer the implied hourly wage by dividing the reported annual salary by 2,080. To make compensation figures comparable across sources, we apply an overhead multiplier of 1.3 *only* when a source reports wages and salaries (*i.e.*, wage-only figures). This adjustment is motivated by the BLS *Employer Costs for Employee Compensation* (ECEC) release for June 2025 (released Sep. 12, 2025): total employer compensation for civilian workers averaged \$48.05 per hour worked, comprising \$33.02 in wages and salaries and \$15.03 in benefits. Benefits therefore account for about 31% of total compensation $\left(\frac{15.03}{48.05} \approx 31\%\right)^4$. For sources that already report total compensation, we do not apply any multiplier.

Table 1: Economic value ($USD) of $OneMillion-Bench by domain and regional subsets.

Domain	**Count**	**Subset**	**Avg. Time (h)**	**Avg. Wage (/h)**	**Avg. Value/Q**	**Total Value**
Finance	80	CN (¥)	23.5	312.6	7,410.8	296,432
		Global ($)	26.1	175.7	4,593.2	183,726
Law	80	CN (¥)	26.2	180.4	4,722.5	188,901
		Global ($)	31.4	243.0	7,666.1	306,646
Healthcare	80	CN (¥)	22.9	189.9	4,379.9	175,197
		Global ($)	22.9	354.9	8,188.9	327,557
Natural Science	80	CN (¥)	22.4	118.5	2,687.3	107,493
		Global ($)	22.4	84.6	1,907.0	76,280
Industry	80	CN (¥)	24.5	152.1	3,845.2	153,809
		Global ($)	24.5	114.9	2,854.0	114,161
Total (Total Value)					CN (¥)	**921,832**
					Global ($)	**1,008,370**

2.2 Expertise Measurement

To evaluate whether generations satisfy expert-defined requirements, we introduce **Expert Score**, a rubric-based measurement to assess the degree to which a generation fulfills professional criteria.

Expert Score. For a question q, the score is defined as

$$\text{Expert Score}(q) = \max\left(0, \frac{\sum\limits_{r \in R_q} s_r}{\sum\limits_{r \in R_q^+} w_r}\right), \tag{1}$$

where R_q denotes the set of rubrics associated with q, R_q^+ is the subset with positive weights, s_r is the score assigned to rubric r, and w_r is its predefined weight. The resulting score is clipped to $[0, 1]$.

[3] https://www.biglawinvestor.com/biglaw-salary-scale
[4] https://www.bls.gov/news.release/ecec.nr0.html

Pass Rate. We also define Pass Rate, a binary success metric indicating whether a generation meets a minimum professional standard via calculating the average Expert Score across all questions Q:

$$\text{Pass Rate}(Q) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}\left[\text{Expert Score}(q) \geq 0.7\right], \tag{2}$$

where $\mathbb{1}[\cdot]$ denotes the indicator function based on the given binary scoring criterion.

Score Aggregation Strategy. Scores are aggregated at multiple levels to analyze agent performance from different perspectives. Domain-level scores are obtained by averaging the question scores within each domain. To examine agentic capabilities across functional dimensions, scores can also be aggregated by rubric type by averaging the normalized rubric scores of the same type across the dataset. Finally, the overall benchmark score is computed by averaging the domain-level scores, providing a balanced estimate of the agent's expertise across domains.

3 Constructing the $OneMillion-Bench

3.1 Data Curation Pipeline

Following the principle: (i) workflow-realistic problems with high economic value, (ii) tasks with objective and quantifiable rubrics, (ii) diverse rubric morphologies without reward hacking, The data curation pipeline goes through a rigorous, multi-expert process three-stage data annotation pipeline as below to ensure the objectivity and professional integrity of the dataset.

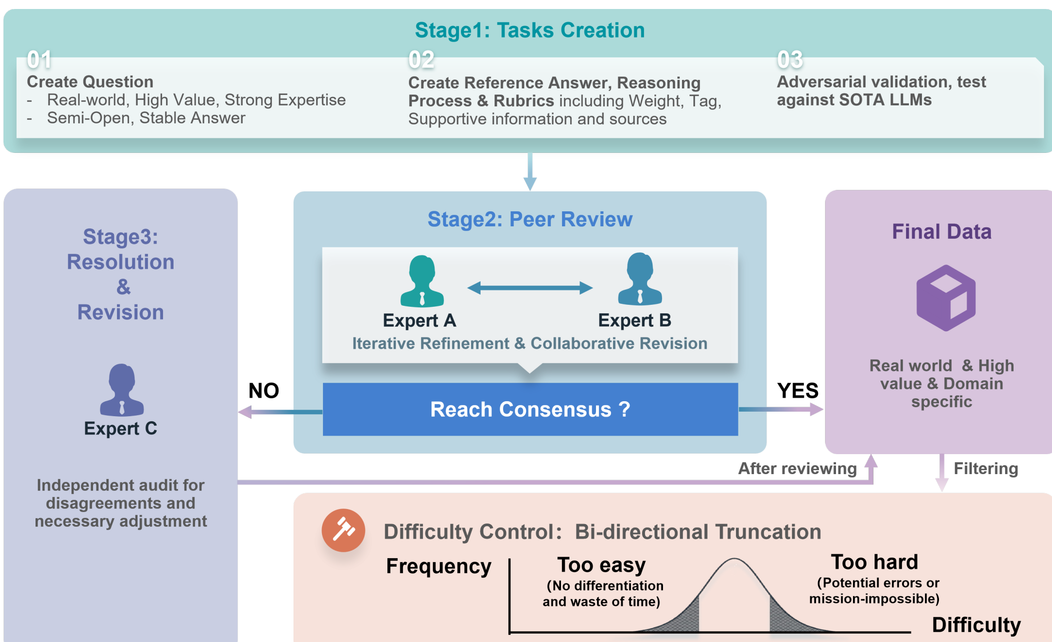

Figure 2: Data curation pipeline of $OneMillion-Bench. The process involves domain experts designing specialized tasks with scoring rubrics, which are then peer-reviewed, validated against state-of-the-art agents to ensure discriminative power, and further refined through consensus.

Stage 1: Task Creation. An expert with deep domain-specific knowledge initiates the process by designing a specialized task. This involves: (1) formulating a specific, practically valuable task; (2) drafting a comprehensive reference answer that illustrates the underlying reasoning process; and (3) defining a detailed corresponding set of scoring rubrics. Tasks are designed to be semi-open-ended with multiple valid approaches and time-invariant with consistent answer over time.

A critical subsequent step is to implement the concurrent adversarial validation. The created task is tested against a series of frontier agents, selected to represent state-of-the-art capability at the time of collection and to reduce single-agent bias by covering different model families. The task is only retained if several agents fail to reach a pre-defined passing threshold according to the experts' rubrics. This validation ensures that the tasks are effectively differentiating between varying levels of agentic capability rather than being trivially solvable.

Stage 2: Peer Review. Upon completion of the initial draft, the task undergoes independent peer review by a second specialist within the same sub-domain. The reviewer performs a holistic evaluation, scrutinizing the task's clarity, domain-specific specialization, alignment with design principles, and the overall fairness of rubrics. Structured feedback is provided, leading to iterative discussions and collaborative revisions between the two experts. This stage continues until a mutual consensus is reached on the task, the reference answer, and the evaluation criteria.

Stage 3: Resolution and Revision. For cases exhibiting with potential risk or instances where the first two experts cannot reach a consensus, a third expert performs an independent audit. This final stage involves necessary adjustments to guarantee the high quality and reliability of the curated data.

To enhance the overall discriminative, we further filtered the candidate dataset with bidirectional truncation by removing or revising tasks at both extremes:

- **Lower Bound Elimination:** Tasks with excessively low difficulty are removed. They are solved by all the agents tested that are deemed as waste of evaluation resources without differentiation.
- **Upper Bound Review:** Tasks with extreme difficulty, where agents consistently yield universally low scores, undergo a secondary review. This crucial step distinguishes between genuinely poor agent performance and undesirable mission-impossible tasks.

Table 2: Data statistics of $OneMillion-Bench

Subset	Domain	# Questions	Question Len.	# Rubrics	# Negative Rubrics
CN	Finance	40	554.8	18.8	4.2
	Law	40	709.1	16.3	3.9
	Healthcare	40	584.2	16.9	4.3
	Natural Science	40	555.8	16.8	4.3
	Industry	40	965.2	15.9	4.0
Global	Finance	40	1810.9	18.5	4.1
	Law	40	3277.8	16.0	4.2
	Healthcare	40	2029.9	16.9	4.2
	Natural Science	40	1634.6	16.8	4.3
	Industry	40	2581.1	15.9	4.0

3.2 Data Overview

Comprehensive Domain Coverage. $OneMillion-Bench comprises a meticulously curated collection of 400 questions categorized into 5 domains including law, finance, healthcare, natural science and industry. The benchmark can be further partitioned into 37 subdomains and 86 third-level categories (see Appendix B). These questions are designed based on the actual needs of different positions, workflows and constraints, which is critical for testing agents in a wide array of real-world scenarios, thereby facilitating a comprehensive and robust evaluation. Examples are depicted in Figure 4 for reference. By simulating real-world industrial workflows, $OneMillion-Bench can deeply examine the language agent's understanding of complex domain knowledge, adherence to professional rules, and its ability to make continuous decisions and executions.

Workflow-realistic Difficulty and High Economic Value Problems. The benchmark focuses on building real-world agentic tasks covering vertical domains, ensuring that the task design closely matches the actual needs and workflows of the industry, thus effectively evaluating the decision-making and task execution capabilities of agents in complex professional environments.

$OneMillion-Bench tasks are constructed to simulate professional workflows which not only require the agent to perform deep reasoning and real-time information integration, but also to clearly

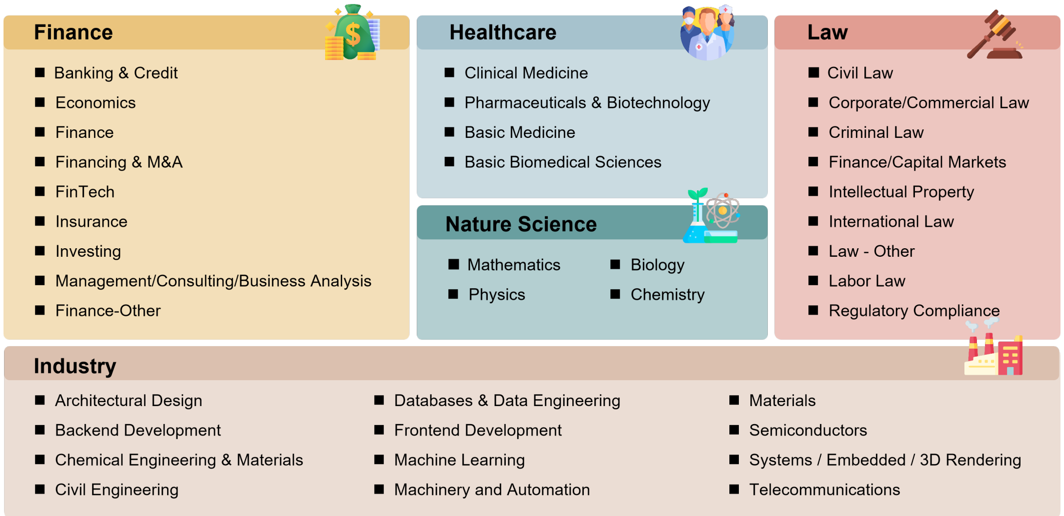

Figure 3: $OneMillion-Bench consists of 5 macro domains, 37 sub-domains and 92 third-level categroies covering a wide variety of real applications and professional scenarios.

distinguish the practical capabilities and economic value of different agents through the accuracy and insight of the agent's generation. In contrast to the exam-style questions of existing benchmarks [7, 29], many tasks in $OneMillion-Bench are intentionally structured so that a correct final answer is inseparable from a correct process: precise retrieval, traceable justification, and careful constraint satisfaction.

Rubrics-based Grading with Negative Penalty. Rubrics-based evaluation has been applied to make transparent the process of synthesising evidence into an overall evaluative judgement, which is *not just a "nice-to-have" but a critical necessity* for validity, reliability, and effective learning for agent improvement. Rubrics in $OneMillion-Bench are primarily determined case-by-case across different domains and hand-crafted by domain experts. It ensures the rubrics are tailored and precise, moving beyond generic scoring to capture domain-specific logic, factual accuracy, and analytical depth. Each effective rubric evaluation consists of four fundamental elements:

- **Rubrics (What to Measure?):** Evaluation criterion, trait and dimension which is to be measured that contributes to correct or incorrect response.
- **Rubrics Weights (What Matters Most?):** Explicit gains / penalties showing the significant of rubrics on evaluation.
- **Rubrics Tags (How to Sort?):** Categorization classifying the criterion per the requirement of different capabilities.
- **Source & Citation:** Citations to official documents, external reference verified for factual faithfulness.

It is worth highlighting that, we propose a **negative rubrics scoring** mechanism to better align with real-world domain requirements. $OneMillion-Bench has asymmetric rubric weights ranging from -20 to 10. These negative rubrics are mostly structured around agent's behaviors including violations of industry-specific norms or professional conduct, unsafe and harmful generation, factual hallucinations and lapses in expected foundational competencies such as instruction following. This design mirrors practical logic and actively steers the evaluation toward domain appropriateness and operational robustness.

Diverse Agentic Capability Assessment. For each rubrics, we affiliate a rubrics tag indicating the agentic capability required to finish the task. The expected capabilities can be divided into 4 classes:

- **Web Search:** It assesses whether a language agent can accurately locate and extract task-related factual information, entity facts, and authoritative definitions from massive, dynamic

Economics & Finance

Background

You are an actuary, currently working in the field of reserve valuation, specifically focusing on the calculation of various statutory reserves and reserves for financial reporting. Currently, due to the transition to IFRS 17, the calculation methods for various reserves have undergone significant changes. Your supervisor wants you to conduct the following research on the calculation of individual life insurance reserves and suggests that you refer to the contents of the fifth edition of the book 'Statutory Valuation of ILA Contracts'.

Question

1. Please list the four methods used by life insurance companies to calculate reserves prior to the advent of the Valuation Manual, and provide a detailed background on the characteristics of these calculation methods regarding assumptions such as premiums and operating expenses.
2. Please indicate which of the aforementioned methods utilize the concept of Expense Allowance. Based on what practical operational characteristics of insurance companies did the concept of Expense Allowance arise? What factors are typically considered in its subsequent amortization adjustment?
3. In practical work, it is often necessary to estimate reserve valuations between years. Using a term life policy as an example, please explain how reserves should be estimated under the assumption of continuous dates, identifying the specific method typically used for this estimation.

Rubrics

Factual Information

+9 List the four reserve calculation methods prior to the advent of the Valuation Manual: Net Level Premium Method (NLP), Full Preliminary Term Method (FPT), Modified Reserve Method, and Commissioners Reserve Valuation Method (CRVM).

+7 Elucidate that the Full Preliminary Term Method (FPT) sets the first-year net premium as the 'net premium for one-year term insurance,' implying that the first year covers only the cost of risk for that year.

+5 Point out that expenses (and lapses) are not explicitly assumed in the Net Level Premium Method (NLP) valuation formula and NLP is calculated solely based on expected future business cash flows.

Analytical Reasoning

+8 Analyze the reason for the proposal of the Expense Allowance concept, namely that the first-year acquisition costs of life insurance contracts are extremely high, necessitating a mechanism for offsetting and amortization.

-4 The model failed to point out that the Force of Interest method is typically used for estimation.

Structure and Formatting

-3 The formulas provided do not use Markdown formula blocks or LaTeX syntax, but instead use plain text or garbled characters, resulting in mathematical symbols that cannot be correctly read.

Law

Background

The UK government, through its intelligence agencies, has ascertained that a cohort of migrants from Northern Nigeria harbors extremist religious views, posing a threat to British society. Intelligence sources further revealed a conspiracy by said migrants to perpetrate terror attacks across the United Kingdom. Acting swiftly, the British Prime Minister persuaded Parliament to establish a Tribunal of Inquiry to investigate the matter. Furthermore, the Prime Minister contacted the President of Nigeria, advising the establishment of a National Tribunal of Inquiry to investigate religious extremism within that country.

Question

Discuss the legal basis for the President of Nigeria to establish a National Tribunal of Inquiry to investigate religious extremism in Nigeria.

Rubrics

Factual Information

+10 Cite the provisions regarding executive powers pursuant to Section 5 of the 1999 Constitution of Nigeria.

+9 Point out that the powers of the President of Nigeria are circumscribed by the legislative scope of the National Assembly.

+8 Point out that the Tribunals of Inquiry Act is deemed an existing law pursuant to Section 315 of the Constitution, yet it lacks constitutional validity at the federal level.

-3 The response includes suggestions regarding administrative engagement by security agencies, which constitutes an irrelevant and extraneous point.

Analytical Reasoning

+6 Argue that the National Assembly lacks the jurisdiction to enact legislation regarding Tribunals of Inquiry applicable to the entire Federation.

-2 The model incorrectly discusses the legal basis from the perspective of international or diplomatic cooperation, rather than strictly based on Nigerian domestic law.

Structure and Formatting

-4 The model fails to point out that the only remedy for establishing a National "Tribunal of Inquiry" is a constitutional amendment, or fails to mention the futility of the National Assembly enacting special laws.

Healthcare & Medicine

Background

Hypothesis: Do specifically enriched bacteria in the gut of Crohn's disease (CD) patients inhibit the expression of the bile acid transporter ASBT via the activation of inflammatory signaling pathways, thereby reducing intestinal bile acid reabsorption and causing luminal bile acid accumulation? Does this subsequently promote Clostridioides difficile spore germination, ultimately exacerbating the incidence and recurrence of CDI?

Question

Please design an experimental protocol to verify this mechanism.

Rubrics

Factual Information

+8 Explicitly propose causal perturbation strategies capable of directly testing the necessity of key nodes, and design clearly controlled functional validation experiments based on these. Specifically, the design must employ loss-of-function approaches targeting ASBT or its upstream inflammatory signaling nodes (e.g., inhibitors or KO mouse models) to compare changes in bile acid transport function or relevant phenotypes pre- and post-perturbation, thereby determining the causal status of the molecule in the hypothesized mechanism. Mere observational detection or correlation analysis does not constitute valid causal verification."

+7 Utilize Inflammatory Bowel Disease models (DSS or TNBS) for mechanistic exploration in animal experiments.

+6 Explicitly define clinical cohort grouping and follow-up design. The model must strictly categorize CD patients into 'CD+CDI' and 'CD' groups, and conduct prospective follow-up on the CDI group to distinguish between recurrent (rCDI) and non-recurrent subgroups.

Analytical Reasoning

+8 The experimental design establishes clear control groups in key mechanistic validation steps to distinguish 'effects mediated by target bacteria or inflammatory signals' from background effects; for example, including non-colonized controls, non-target bacteria controls, or non-induced inflammation controls, ensuring that changes in ASBT expression and relevant phenotypes are attributable.

-6 The experimental design fails to detect key molecular indicators (e.g., ASBT protein/mRNA expression, intestinal bile acid concentration, inflammatory cytokines), or relies solely on macroscopic phenotypic observations (e.g., recurrence rate) without molecular-level validation.

Structure and Formatting

-7 The experimental protocol lacks verification of intermediate mechanistic steps (e.g., omitting detection of ASBT expression, bile acid reabsorption, or spore germination), directly linking microbiota changes to CDI recurrence (functional outcome).

-9 Substitutes specific mechanistic action with macroscopic disease or ecological background descriptions.

Figure 4: Sample data of different domains with varying scores of rubric weights and tags.

information sources, and ensures their accuracy and completeness. Entities, definitions, and references in the response are evaluated for accuracy and completeness. Acquiring objective data usually relies on the intrinsic information retrieval capability of agents for fact seeking and verfication focusing on "what" question.

- **Reasoning:** Beyond searching factual information, it focus on how language agents make reasonable causal attributions, coherent trend judgments, and in-depth logical deduction and insights (such as attribution analysis and trend interpretation), corresponding to solving the questions of "why" and "how".
- **Verbalization:** It focuses on the logical flow, organizational structure, language style, and overall readability of the response from the agent in a clear, professional and context-appropriate manner ensuring effective communication.
- **Instruction Following:** Instruction compliance is crucial for the agent's reliability in tasks from professional domains. It rigorously evaluates the agent's adherence to explicit constraints, rules and guidance when applying professional knowledge and executions. It ensures that the agent's behavior is controllable, predictable, and conforms to the user's specific requirements.

Bilinguality Integrated with Local Culture. While numerous general-purpose benchmarks exist, they often operate under a default English-centric or western-centric context, failing to capture significant variations in tasks, regulations, and practical scenarios encountered in different linguistic and cultural environments. Therefore, our benchmark consists of 200 English and 200 Chinese instances. The Chinese set, however, is not a direct translation but a purpose-built collection that incorporates tasks specific to the context of Mainland China. This includes local regulations and industry standards (*e.g.*, China's Cybersecurity Law, healthcare insurance policies, Chinese accounting standards) and cultural nuances in scenario-centric problems. The benchmark explicitly tests for this contextual adaptability, measuring an agent's ability to operate not just with general domain knowledge, but with the precise, localized knowledge required for practical application.

4 Benchmarking Frontier Agents on $OneMillion-Bench

4.1 Evaluated Systems

In this section, we present the evaluation results for $OneMillion-Bench, encompassing a total of 35 models, grouped into three categories according to their characteristics:

- **Vanilla Models (17 models):** The vanilla models represent open-source and proprietary models evaluated without external tool usage, including Claude-Opus-4.6 [1], DeepSeek-V3.2-Speciale [14], Doubao-Seed-2.0-Pro [5], Gemini-3-Pro-Preview [8], Gemini-3.1-Pro-Preview[34], GLM-5 [41], GPT-5.2-High [23], GPT-5.3-Codex [24], GPT-5.4-High [25], Grok-4 [37], Hunyuan-2.0-Thinking [33], Kimi-K2.5 [20], Ling-2.5-1T [10], MiniMax-M2.1 [19], MiniMax-M2.5 [18], Qwen3.5-Plus [27], and Step-3.5-Flash [31].
- **Search Agents (vanilla models with web search tools, 17 agents):** This category evaluates all vanilla models listed above equipped with web search tool calling for real-time information retrieval, *e.g.*, Claude-Opus-4.6 (Web Search) [1].
- **Deep Research Agents (3 agents):** These are specialized systems optimized for complex reasoning and long-context research tasks, including o3-DeepResearch [22], o4-Mini-DeepResearch [22], Sonar-DeepResearch [26].

4.2 Main Results

Tables 3, 4, 7, and 8 present the comparison for all evaluated models on the Global and CN subsets, respectively. From these empirical results, several key observations emerge.

A Clear Leader Emerges, and the Gap Widens with Web Search. In both languages, CLAUDE-OPUS-4.6 achieves the best overall performance among vanilla models and remains the top performer with search enabled capability. Notably, the search-enabled setting amplifies the advantage: both the overall Expert Score and Pass Rate in the top row, indicating that a high Expert Score is not merely

Table 3: Comparison of Economic Value ($USD), Expert Score (%), and Pass Rate(%) on **Global set**.

Model / Agentic System	Economic Value		Expert Score		Pass Rate	
	Vanilla	Search	Vanilla	Search	Vanilla	Search
CLAUDE-OPUS-4.6	439.2k	483.8k ↑44.6	55.0	63.0 ↑8.1	36.5	43.5 ↑7.0
GPT-5.4-HIGH	330.9k	365.5k ↑34.6	55.0	59.2 ↑4.2	31.5	38.0 ↑6.5
QWEN3.5-PLUS	256.0k	263.7k ↑7.8	47.6	49.0 ↑1.4	22.5	23.5 ↑1.0
GPT-5.2-HIGH	236.6k	335.9k ↑99.3	47.6	56.9 ↑9.3	23.0	32.0 ↑9.0
GEMINI-3.1-PRO-PREVIEW	207.5k	245.7k ↑38.2	47.9	44.6 ↓3.4	18.5	21.0 ↑2.5
KIMI-K2.5	203.3k	162.7k ↓40.6	45.1	41.2 ↓3.9	17.5	17.5 ↑0.0
GPT-5.3-CODEX	201.1k	317.8k ↑116.7	48.1	50.6 ↑2.6	18.5	29.0 ↑10.5
GEMINI-3-PRO-PREVIEW	187.3k	345.3k ↑158.0	43.7	52.8 ↑9.1	15.5	28.5 ↑13.0
DEEPSEEK-V3.2-SPECIALE	154.9k	118.0k ↓37.0	40.5	38.6 ↓1.8	16.0	13.0 ↓3.0
GROK-4	136.3k	204.6k ↑68.3	38.6	45.9 ↑7.2	12.0	18.5 ↑6.5
DOUBAO-SEED-2.0-PRO	123.9k	330.7k ↑206.9	39.4	51.8 ↑12.4	13.5	28.5 ↑15.0
GLM-5	107.8k	177.9k ↑70.1	38.5	41.1 ↑2.6	9.5	16.5 ↑7.0
MINIMAX-M2.1	85.6k	46.6k ↓39.0	34.9	33.9 ↓1.0	8.0	5.5 ↓2.5
LING-2.5-1T	79.5k	50.0k ↓29.5	33.1	31.2 ↓1.9	7.5	6.5 ↓1.0
STEP-3.5-FLASH	68.6k	46.9k ↓21.7	34.4	33.0 ↓1.4	7.0	5.0 ↓2.0
HUNYUAN-2.0-THINKING	68.0k	25.6k ↓42.3	34.7	30.2 ↓4.5	8.5	3.0 ↓5.5
MINIMAX-M2.5	37.9k	82.6k ↑44.8	29.5	31.5 ↑2.0	4.0	8.0 ↑4.0
Deep Research Agents						
O3-DEEPRESEARCH	181.7k		46.3		17.0	
SONAR-DEEPRESEARCH	122.8k		42.3		13.5	
O4-MINI-DEEPRESEARCH	112.6k		37.7		10.5	

a result of covering isolated aspects of the rubric, but also corresponds to a greater proportion of questions surpassing the passing threshold.

Web Search is not Always Beneficial. While search improves Expert Score/Pass Rate for top models (*e.g.*, CLAUDE-OPUS-4.6), we observe search-induced regressions for several systems. For instance, HUNYUAN-2.0 drops from $34.7 \rightarrow 30.2$ in Expert Score and $8.5\% \rightarrow 3.0\%$ in Pass Rate on Global, and from $36.3 \rightarrow 32.8$ with $9.5\% \rightarrow 6.5\%$ on CN; STEP-3.5-FLASH also slightly decreases (Global: $34.4 \rightarrow 33.0$; CN: $36.2 \rightarrow 35.6$). This indicates that retrieval can introduce noisy or conflicting evidence and hurt rubric satisfaction when agents are not robust in identifying evidence.

Deep Research Agents do not Dominate Strong Search-enabled Generalists. Specialized deep research agents (*e.g.*, O3-DEEPRESEARCH, SONAR-DEEPRESEARCH) achieve competitive mid-tier Expert Score, but they generally lag behind the best search-enabled generalist models in overall Expert Score, Pass Rate, and Economic Value on both Global and CN. This indicates that, under rubric-based expertise integrity evaluation, the decisive factor may be *robust rubric coverage and compliance* rather than longer or more complex research-style pipelines.

"Near-miss" appears for Pass Rate vs. Expert Score. While many models reach moderate Expert Score (*e.g.*, ∼45-50% on Global/CN), their Pass Rates can remain much lower (often below ∼25%), indicating that performance is frequently distributed as *partially satisfying many rubrics* rather than *fully meeting a competence threshold* on a substantial portion of questions. Therefore, Pass Rate complements Expert Score by exposing whether improvements reflect broad but shallow gains or genuinely push examples over the acceptance boundary (Expert Score$(q) \geq 0.7$).

Domain Difficulty is Non-uniform and Persists across Languages. Performance varies markedly by domain: Finance tends to be consistently challenging (lower Expert Score across many models), whereas Healthcare/Law often yield higher scores for top systems. Importantly, the domain profile is broadly stable between Global and CN, suggesting that the benchmark difficulty is driven not only by language factors but also by domain-specific rubric strictness and knowledge requirements.

Table 4: Comparison of Economic Value (¥CNY), Expert Score (%), and Pass Rate (%) on **CN set**.

Model / Agentic System	Economic Value		Expert Score		Pass Rate	
	Vanilla	Search	Vanilla	Search	Vanilla	Search
Claude-Opus-4.6	350.3k	470.2k ↑119.9	55.8	64.5 ↑8.7	35.0	48.5 ↑13.5
GPT-5.4-High	339.5k	305.4k ↓34.1	56.2	58.7 ↑2.5	35.5	35.5 ↑0.0
Qwen3.5-Plus	304.9k	308.1k ↑3.2	51.4	52.1 ↑0.6	28.0	28.0 ↑0.0
Gemini-3.1-Pro-Preview	264.1k	201.5k ↓62.6	49.0	46.9 ↓2.1	23.0	19.5 ↓3.5
GPT-5.2-High	226.9k	294.0k ↑67.1	47.1	52.6 ↑5.5	22.0	28.0 ↑6.0
Gemini-3-Pro-Preview	202.9k	332.5k ↑129.7	46.0	54.0 ↑8.0	20.0	30.0 ↑10.0
Kimi-K2.5	196.5k	126.0k ↓70.4	45.4	43.7 ↓1.7	19.5	15.0 ↓4.5
GPT-5.3-Codex	178.4k	201.1k ↑22.7	46.9	50.9 ↑4.1	19.0	23.0 ↑4.0
DeepSeek-V3.2-Speciale	139.4k	155.8k ↑16.4	41.1	44.4 ↑3.3	14.0	16.5 ↑2.5
GLM-5	138.9k	136.3k ↓2.6	40.7	41.1 ↑0.4	13.0	11.0 ↓2.0
Ling-2.5-1T	119.8k	101.8k ↓17.9	36.7	35.6 ↓1.1	11.0	11.5 ↑0.5
Doubao-Seed-2.0-Pro	110.8k	255.9k ↑145.1	40.6	50.5 ↑9.8	11.5	25.0 ↑13.5
MiniMax-M2.1	106.6k	67.2k ↓39.4	35.8	36.7 ↑1.0	10.0	6.0 ↓4.0
Grok-4	104.2k	99.4k ↓4.8	39.8	39.6 ↓0.2	11.0	11.5 ↑0.5
Hunyuan-2.0-Thinking	83.0k	53.8k ↓29.2	36.3	32.8 ↓3.4	9.5	6.5 ↓3.0
MiniMax-M2.5	70.3k	51.7k ↓18.7	33.4	32.1 ↓1.3	7.5	7.5 ↑0.0
Step-3.5-Flash	52.6k	68.7k ↑16.2	36.2	35.6 ↓0.6	6.5	6.5 ↑0.0
Deep Research Agents						
o3-DeepResearch	149.2k		41.7		16.0	
Sonar-DeepResearch	129.5k		40.2		13.5	
o4-Mini-DeepResearch	84.7k		36.0		9.0	

Language Effects are Model-dependent, Suggesting Various Proficiency of Bilingual Robustness. Comparing Global vs. CN, some model families maintain similar Expert Score (*e.g.*, top-tier systems remain stable), while others show larger shifts, implying heterogeneous cross-lingual generalization. This model-dependent variation further motivates reporting both Global and CN results: it helps disentangle *knowledge/rubric competence* from *language-specific execution quality*.

4.3 Performance Analyses across Rubric Type

To evaluate task performance in detail, we report the per-type scores in Table 5 and observe that:

Rubric-type Performance is Highly Non-uniform. Aggregated over Global/CN, models generally achieve the highest scores on *Structure and Formatting* and *Instructions Following*, while *Factual Information* and *Analytical Reasoning* remain more challenging. This gap suggests that many systems can produce well-structured outputs and follow surface constraints, yet still struggle to consistently provide correct domain facts or complete multi-step analyses under expert rubrics.

Web Search Benefits Evidence-centric Rubrics. Across models, Search brings the most consistent gains to *Factual Information* and *Analytical Reasoning*, especially for strong systems (e.g., Claude-Opus-4.6, Qwen3-Max, GPT-5.2). This suggests that external retrieval primarily improves evidence coverage and verifiability. However, weaker models sometimes see factual or reasoning drops, indicating that retrieval quality and integration remain critical.

Web Search Acts as a Efficacy Amplifier. A clear stratification emerges: strong models tend to improve across multiple rubrics under Search, while weaker models often degrade, particularly in reasoning and formatting. This implies that Search amplifies underlying capabilities—models with better evidence filtering and planning can convert retrieval into score gains, whereas weaker models are more vulnerable to noise.

Instruction Following is the Most Fragile Ability. *Instructions Following* shows sharp polarization. Some strong models gain substantially under Search, while others decline notably. This suggests that

Table 5: Expert Score aggregated by rubric type, averaged over Global and CN subsets. FI: Factual Information. AR: Analytical Reasoning. IF: Instructions Following. SF: Structure and Formatting.

Model / Agentic System	FI		AR		IF		SF	
	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search
GPT-5.2-High	47.4	59.4 ↑12.0	52.3	56.4 ↑4.1	66.0	72.1 ↑6.1	82.7	85.6 ↑2.9
Qwen3.5-Plus	50.3	50.0 ↓0.3	58.9	59.9 ↑1.0	73.2	74.6 ↑1.4	86.4	88.8 ↑2.4
Claude-Opus-4.6	54.0	66.5 ↑12.5	61.3	73.0 ↑11.7	74.6	82.1 ↑7.5	88.9	90.0 ↑1.1
GPT-5.4-High	53.5	63.5 ↑10.0	60.8	61.6 ↑0.8	71.5	73.9 ↑2.4	86.4	84.5 ↓1.9
Gemini-3.1-Pro-Preview	49.8	46.7 ↓3.1	51.9	50.8 ↓1.1	70.6	67.9 ↓2.7	82.7	82.7 ↑0.0
GPT-5.3-Codex	48.5	58.4 ↑9.9	51.1	52.0 ↑0.9	67.5	65.9 ↓1.6	85.0	83.6 ↓1.4
Kimi-K2.5	46.5	44.1 ↓2.4	51.1	48.1 ↓3.0	70.1	63.4 ↓6.7	85.0	80.6 ↓4.4
Gemini-3-Pro-Preview	45.8	56.7 ↑10.9	49.2	60.7 ↑11.5	66.9	73.1 ↑6.2	82.8	86.8 ↑4.0
Grok-4	41.6	48.5 ↑6.9	44.7	46.7 ↑2.0	65.2	62.7 ↓2.5	83.0	86.5 ↑3.5
Doubao-Seed-2.0-Pro	42.3	53.6 ↑11.3	45.3	58.8 ↑13.5	62.3	72.7 ↑10.4	83.6	86.8 ↑3.2
DeepSeek-V3.2-Speciale	43.7	43.9 ↑0.2	45.4	46.8 ↑1.4	62.8	60.2 ↓2.6	79.4	78.8 ↓0.6
GLM-5	41.8	40.8 ↓1.0	45.1	47.7 ↑2.6	65.3	65.7 ↑0.4	79.4	82.7 ↑3.3
Hunyuan-2.0-Thinking	39.1	36.3 ↓2.8	40.2	36.5 ↓3.7	63.5	60.9 ↓2.6	85.0	79.8 ↓5.2
MiniMax-M2.1	39.3	38.3 ↓1.0	41.6	42.4 ↑0.8	60.4	57.4 ↓3.0	82.6	81.8 ↓0.8
Step-3.5-Flash	36.2	38.4 ↑2.2	42.5	41.6 ↓0.9	62.3	56.1 ↓6.2	82.6	79.2 ↓3.4
Ling-2.5-1T	37.6	36.5 ↓1.1	41.6	38.7 ↓2.9	59.3	55.9 ↓3.4	79.3	74.4 ↓4.9
MiniMax-M2.5	34.9	35.3 ↑0.4	39.1	35.8 ↓3.3	58.2	55.4 ↓2.8	83.0	71.5 ↓11.5
Deep Research Agents								
Sonar-DeepResearch	47.2		49.9		60.7		80.7	
o3-DeepResearch	47.9		50.2		59.2		76.5	
o4-Mini-DeepResearch	42.9		42.3		58.1		71.7	

using tools under constraints requires robust planning and compliance control; otherwise, injected evidence may cause deviation from task requirements.

Formatting Reflects Integration Quality. *Structure and Formatting* scores are already high at baseline, leaving limited headroom. Search can easily destabilize formatting due to longer, citation-heavy responses. Models that still improve likely possess stronger response planning and structured generation abilities, making formatting a proxy for retrieval integration quality.

Deep Research Agents are Competitive but not Dominant. Dedicated deep research agents achieve solid pass rate, but they do not outperform the strongest vanilla models with Search enabled.

4.4 Impact of Web Search Scaffolds

Agentic scaffolds have become the de facto foundation that affects task performance. Specifically, scaffolds provide specialized tools, structured prompts for tool calling, and controlled environments that enable agents to retrieve and leverage external knowledge. To mitigate non-transferability issues in the evaluation and provide further insights into the impact of agentic scaffolds on $OneMillion-Bench, we investigate this question with three setups: (1) official provider scaffolds (*e.g.*, OpenAI and Anthropic), (2) OpenRouter, and (3) a setting without scaffolds (w/o web search). As shown in Figure 5, official scaffolds outperform OpenRouter in almost all evaluated models. The performance gap between official and OpenRouter scaffolds varies considerably across models — Claude-Opus-4.6 and Gemini-3-Pro-preview exhibit relatively larger gaps, while Kimi-K2.5 shows comparable performance across both scaffolds. Furthermore, the "w/o Search" condition reveals that web search is a critical component for most models, as removing it generally degrades performance, though in some cases (*e.g.*, Claude-Opus-4.6) the "w/o Search" baseline surprisingly surpasses the OpenRouter setup, suggesting that scaffold quality can matter more than access to web search alone.

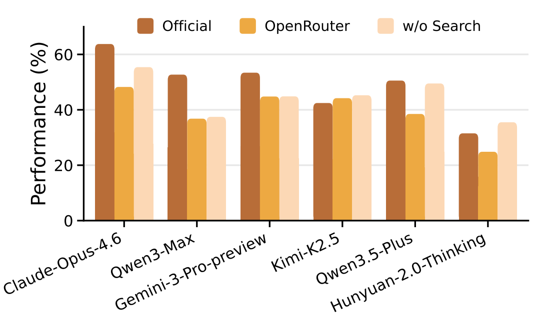

Figure 5: Comparison of web search scaffolds.

4.5 Sensitivity Analysis on LM-as-a-judge

The reliability of LM-as-a-judge evaluation hinges on the judge's inherent inductive biases, which may potentially favor certain response styles. To assess this, we assess five representative agents with six various LM judges (*c.f.*, Figure 6). We find the partial order remains moderately consistent across judges, suggesting robustness to individual judge idiosyncrasies. Nevertheless, notable rigor differences emerge: GPT-5.2-High is the most strict judge while GLM-5 the most lenient, with the gap most pronounced for models like Claude-Opus-4.6 ($\sim$8%) and narrowing for weaker ones, indicating stricter judges are more discriminative at the top of the performance range. Crucially, agent rankings remain stable across all judges, though absolute scores vary with judge strictness, underscoring the need to report judge identity and adopt multi-judge evaluation to mitigate inductive bias.

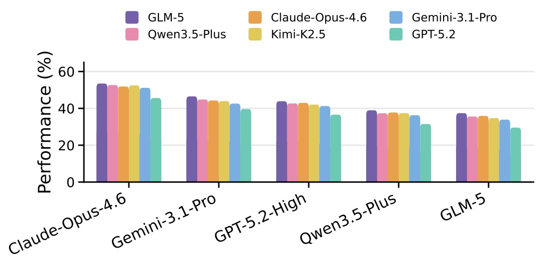

Figure 6: Comparison of various grading judges.

4.6 Diachronical Analysis

As depicted in Figure 7, all models consistently achieve higher performance on temporally agnostic questions than on weakly sensitive and sensitive ones, revealing that temporal sensitivity poses a genuine and progressive challenge. The performance drop from agnostic to sensitive is most pronounced for Claude-Opus-4.6 and GPT-5.2-High ($\sim$15% to 20%), suggesting these models may over-rely on temporal anchoring cues when reasoning. Notably, Qwen3.5-Plus and GLM-5 exhibit narrower gaps across categories, but at the cost of lower overall performance, indicating limited temporal reasoning capacity rather than robustness. These findings suggest that strong performance and resilience to temporal sensitivity are not always correlated — a nuance worth considering when selecting agents for real-world, time-evolving information tasks.

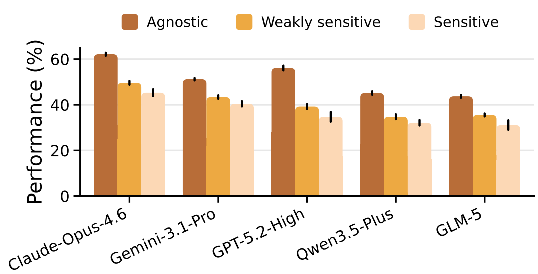

Figure 7: Diachronical analyses (*i.e.*, time-sensitivity analyses) on $OneMillion-Bench.

4.7 Test-time Scalability

We report pass@k and pass^k scores with performing parallel test-time scaling experiments on the Finance subset of the Global split. Figure 8 illustrates the test-time scalability of various models on $OneMillion-Bench. As k increases, the two metrics diverge sharply: pass@k (dotted lines) shows consistent logarithmic gains across all models, with Claude-Opus-4.6 leading and plateauing near 30%, while pass^k (solid lines) decays towards zero. This indicates that larger sample sizes increase the probability of generating at least one correct solution, but introduce variance that degrades aggregated output reliability. Thus, while models such as Gemini-3.0-Pro-Preview and GPT-5.2 scale well in raw capability, they struggle to maintain consistency under high uncertainty with large-k regimes.

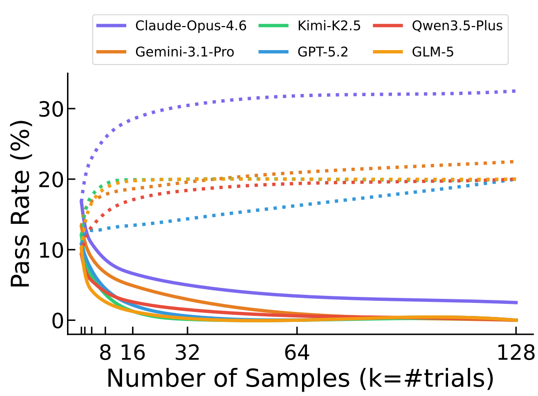

Figure 8: Test-time scalibility on the Finance subset of Global split for reliability demonstration.

5 Discussions

5.1 Pareto Optimality of Return on Cost

Comprehensively, we demonstrate that models achieve a Pareto-optimal trade-off tendency between the outcome economic value versus the inference cost of agents or models, behaving on $OneMillion-Bench. As depicted in Figure 9, it can be seen that specialized search agents deliver drastically higher economic value than that of the same base model. The wide gap between marginal cost and aggregate economic output indicates that agents with search tools capture extremely high profit margins, as they address unmet high-value demand from professional users, enterprises, and researchers who need complex problem-solving and information seeking capabilities rather than basic text generation. We also notice that smaller model deliver near-frontier deep research performance at a fraction of the cost of the largest top-tier models, making advanced research tools accessible to price-sensitive users in the real scenarios. Based on our further case studies in Section 5.4, although there are wide variations in interaction and generation patterns among various models, our analysis shows that higher token cost does not necessarily contribute to better performance in those domain specific tasks.

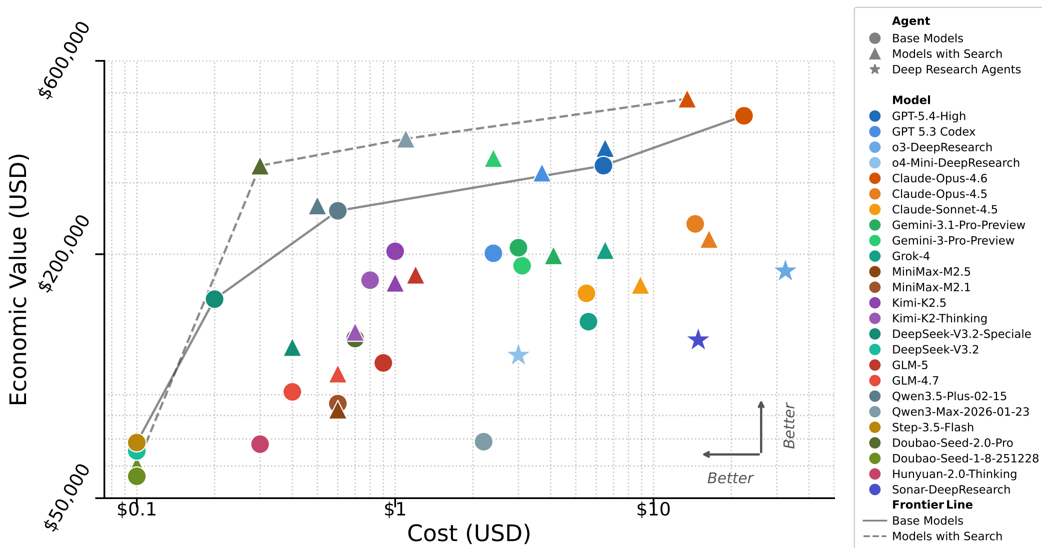

Figure 9: Pareto frontiers of base models, search agents, and deep research agents performance show the real-world tradeoff between economic value and corresponding cost on $OneMillion-Bench. Cost-matched search agents Pareto-dominate most base models using the same amount of overhead.

5.2 Dynamic Expansion and Scenario Deepening

While the current five domains are representative, other numerous high-value fields (*e.g.*, energy, climate science, public policy) remain uncovered. Future work could incrementally incorporate more vertical domains and further refine sub-domains to enhance the benchmark's comprehensiveness and granularity. Beyond, building a live, evolving benchmark with the dynamic nature of real-world expertise by integrating real-time or frequently updated information will be the trend in the long run.

5.3 Automation of Fine-grained Process Evaluation

Although we improve rubric reliability through strict guidelines, verification, and consistency with human review, current rubrics are less objective than the equivalency checker of a single expression or number, and relies on the model judge's capabilities. Furthermore, although structured rubrics designed by experts are employed, full reliance on manual scoring is difficult to scale. Future research could develop more advanced automated evaluation mechanisms, such as rule-based or learning-based methods to automatically score intermediate steps like reasoning chains, evidence citation, and compliance checks, thereby improving efficiency while maintaining reliability.

As a starting point, $OneMillion-Bench aims to propel language agents towards greater professionalism, reliability, and practical utility. We hope $OneMillion-Bench can serve as a practical, challenging evaluation target for the next generation of language agents — systems expected not only to answer, but to *do the work* necessary to answer well. We look forward to collaborating with the community to iteratively improve the benchmark itself and to deeply explore the open questions outlined above, ultimately contributing to the safe, effective, and responsible deployment of AI in critical professional domains.

5.4 Case Studies of Failure Patterns

We collect and identify the underlying cause of failure cases picked from five macro domains in $OneMillion-Bench enabling further improvement.

5.4.1 Web Search is a Double-edged Sword

For questions primarily testing analytical reasoning, web search may introduce outdated, non-authoritative, or loosely related information that disrupts argument structure and causal chains. In *question 6466* (Eco. & Finance.), for instance, the core requirement is professional mechanistic reasoning and contextual comparison rather than fact lookup, and search-enabled models showed noticeable score drops.

By contrast, in Healthcare & Medicine, search improves performance when results directly supply rubric-critical details. However, it proves harmful when retrieved content pulls the agent toward an incompatible guideline system (*e.g.*, *question 9952* requires CSCO conventions, yet retrieval nudges responses toward NCCN or ESMO), or when noisy retrieval erodes required hierarchical structure (*e.g.*, *question 9287*, where the distinction between primary and secondary screening becomes blurred). Overall, web search is a conditional gain rather than a default advantage, it is beneficial for knowledge-completion tasks, but potentially harmful for reasoning- or structure-sensitive ones.

5.4.2 Bad Behaviours in Structural Extraction and Calculation

Economics and Finance questions require precise extraction of figures from financial statements and consistent multi-step derivations. Models frequently exhibit arithmetic errors, unsupported numbers, or omission of key metrics. In *question 8708*, the task demands cross-company comparison of changes in turnover days, earnings quality, and the cash conversion cycle; yet agents struggle to accurately extract and calculate inventory turnover days, accounts receivable turnover days. Such errors sharply distinguish superficial workflow fluency from genuine quantitative competence.

5.4.3 Deficiency in Domain-specific Knowledge and Rule Application

In financial compliance and legal practice settings, correct answers often hinge on recent or highly specific normative documents and cases. For example, *question 914* (Eco. & Fin.) requires citing salient supervisory rules and conducting itemized compliance checks on financial investments as of a specified cutoff date. *Question 1675* (Law) demands correct application of Supreme People's Court Guiding Case No. 226 (issued in 2024), including its decisive reasoning. *Question 1248* involves charge concurrence and punishment doctrine, requiring coherent multi-article statutory reasoning.

Across agents, the common failure is not complete ignorance but imprecise mapping from fact patterns to relevant provisions, case holdings, and local standards, followed by unstable legal qualification. This is particularly salient when local thresholds matter (*e.g.*, regional definitions of a "large amount" in extortion cases). These failures point to a combined bottleneck in normative coverage, pinpoint retrieval, and faithful rule application.

5.4.4 Bottlenack in Complex Reasoning and Exectuable Problem-solving

Despite their ability to produce coherent text, language agents still struggle with tasks requiring deep understanding, multi-step deduction, and long-horizon exploration, often lacking depth and accuracy. They frequently point in the right direction but fall short on actionable, in-depth details. This is especially evident in software engineering, where agents struggle with inherently unsolvable problems or tasks requiring exhaustive possibility exploration. For example, in question 10631, generated test cases suffer from insufficient coverage, failing to handle edge cases and boundary scenarios properly. In machine learning tasks, agents rarely adopt systematic approaches, skipping comprehensive diagnosis of the given context and intermediate findings. While agents can retrieve and chain conceptually related terms from training data, they fail to perform the contextual adaptation and causal analysis required for practical problem-solving (*question 4039*). This likely contributes to performance collapse on high-complexity, real-world tasks demanding multi-step reasoning.

In Healthcare & Medicine, rubrics emphasize actionable clinical elements (*e.g.*, diagnosis, staging, treatment choice, follow-up, and contraindications), yet models frequently omit operational specifics. For question 7240, expected answer points include disease staging of both eyes, follow-up intervals, laser indications, systemic management, and patient education — details that most models miss, resulting in low scores. In the Natural Sciences, Chinese and English prompts yield comparable performance, and enabling search provides only marginal gains, suggesting these questions demand expert analysis over information retrieval. Question weaknesses nonetheless persist: limited anticipation of experimental conditions (question 6426, Physics), insufficiently constrained chemical reasoning (question 6171, Chemistry), and shallow mechanistic understanding (question 6681, Biology).

6 Related Work

Recent benchmarking efforts reveal both substantial progress and persistent gaps in evaluating language agents beyond surface-level correctness.

Hard Question Answering. A line of work focuses on pushing the difficulty of static question answering benchmarks. GPQA introduces expert-written, Google-proof questions designed to require deep domain knowledge and multi-step reasoning [29]. To reduce data contamination and memorization, LiveBench adopts a continuously refreshed evaluation pipeline with newly generated reasoning problems [36]. MMLU-Pro extends the original MMLU by increasing the proportion of reasoning-focused questions and enlarging the answer choice space, discouraging shortcut guessing [9, 35]. More recently, Humanity's Last Exam (HLE) further raises the difficulty using frontier-level academic questions crafted by domain experts [7]. These benchmarks effectively probe the upper limits of model knowledge and reasoning under extreme difficulty. However, they largely evaluate isolated question answering, without modeling the multi-step workflows, evidence reconciliation, and decision justification characteristic of real professional practice.

Agentic Workflow Benchmarks. Another line of work evaluates models in task-oriented agent environments. XBench targets profession-aligned productivity tasks [6], while SWE-bench (Verified) measures software engineering capability through real GitHub issue resolution [12]. Planning-focused benchmarks such as TravelPlanner stress multi-constraint decision making [38], and τ-bench evaluates tool–agent–user interactions under policy constraints [3, 39]. Execution-oriented environments including Terminal-Bench [32] and AndroidWorld [28] further test grounded interaction with real systems. Additional benchmarks such as Vending-Bench and FutureX examine long-horizon coherence and future prediction in dynamic settings [2, 42]. Together, these benchmarks mark a shift from static QA toward agentic execution. However, they primarily measure task completion, while providing limited constraints on the epistemic quality, traceability, and reliability of the underlying reasoning process.

Reality-Grounded Evaluation. A complementary direction evaluates agents in real-world environments where outcomes are determined by external reality. LiveTradeBench formalizes this paradigm using streaming financial markets, showing that leaderboard performance does not reliably translate into real trading outcomes [40]. Public competitions such as Alpha Arena further demonstrate the high variance and regime sensitivity of agent performance in dynamic markets [21]. Beyond finance, Lab-Bench evaluates agents in real scientific workflows involving experimental data and literature analysis [13]. While these settings offer strong ecological validity, they sacrifice diagnostic resolution: failures are difficult to attribute to specific reasoning, retrieval, or decision-making errors.

As a result, $OneMillion-Bench occupies a middle ground between static exam-style benchmarks and unconstrained real-world deployment, enabling more reliable and economically meaningful evaluation of language agents in professional settings.

7 Conclusion

In this work, we present $OneMillion-Bench, a benchmark designed to bridge the critical divide between static, exam-style evaluations and the unconstrained demands of high-stakes professional deployment. By integrating expert-curated workflows with a rubric-based assessment of logical coherence, factual grounding, and professional compliance, $OneMillion-Bench establishes a rigorous standard for evaluating the economic readiness of language agents. Our findings highlight a significant reliability gap, demonstrating that current models often fail to maintain the consistency and evidence-grounding required for autonomous professional labor. Ultimately, $OneMillion-Bench shifts the evaluative focus from mere surface-level correctness toward a framework that prioritizes grounded, compliant, and economically consequential decision-making as the true metric for agentic maturity.

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Appendix

Table of Contents

A Authors

A.1 Data Contributors

We sincerely appreciate all experts for their invaluable data and feedback in advancing AI. Listed below are major contributors who agreed to be named (alphabetical by surname):

Mingyang Che, Jiarui Chen, Hanqi Li, Muzhen Liu, Hang Lu, Pan Wang, Shuo Wang, Wenzhou Wang, Chenyi Xie, Zhenjie Ye, Xia Yue, Jiayi Zhang, Yanru Zhang, Yongmou Zhao, Zishen Zhou.

B Category distributions of $OneMillion-Bench

We identified a total of **92** unique third-level subdomain tags in $OneMillion-Bench in five high-level domains. The non-synonymous tags across languages are kept as separate entries

Healthcare and Medicine: 25 tags.

- Cell and Gene Therapy (CGT)
- Gastrointestinal Surgery
- Pathophysiology
- Hepatobiliary and Pancreatic Surgery
- Cardiothoracic Surgery
- Thyroid and Breast Surgery
- Pathology
- Obstetrics and Gynecology
- Rheumatology and Immunology
- Biopharmaceuticals (Macromolecules)
- Orthopedics
- Molecular Biology
- Ophthalmology
- Medical Genetics
- Dentistry
- Emergency and Critical Care Medicine
- Cardiovascular Medicine
- Neurology
- Medical Laboratory Science
- Respiratory Medicine
- Immunology
- Endocrinology and Metabolic Diseases
- Oncology
- Nephrology
- Neurosurgery

Industry: 12 tags.

- Semiconductors
- Telecommunications
- Machine Learning
- Backend Development
- Databases and Data Engineering
- Civil Engineering
- Chemical Engineering and Materials
- Architectural Design
- Systems / Embedded / 3D Rendering
- Frontend Development
- Machinery and Automation
- Materials

Law: 19 tags.

- Criminal Defense
- Contract Disputes
- Restructuring / Financing / M&A
- Corporate Governance
- Data Compliance/Cybersecurity
- Financial Regulation
- Government Regulation
- Copyright
- Law-Other
- Trademark
- Patent
- Labor Law
- Marriage/Inheritance
- Criminal Compliance
- Securities & Listing (IPO)
- Funds & Asset Management
- Public International Law
- Private International Law
- Tort Disputes

Economics and Finance: 18 tags.

- Equities
- Bonds
- Mergers & Acquisitions
- Management/Consulting/Business Analysis
- Risk Management
- VC/PE
- Accounting
- Investing-Other

- Equity Financing
- Commodities
- Consumer Finance
- Corporate Banking
- Other
- Cryptocurrency
- Derivatives
- Macroeconomics
- Life Insurance
- Quantitative

Natural Sciences: 18 tags.

- Condensed Matter Physics
- Physics-Other
- Organic Chemistry
- Inorganic Chemistry
- Materials Chemistry
- Physical and Theoretical Chemistry
- Classical Physics
- Quantum Physics
- Applied Mathematics
- Mathematics-Other
- Microbiology
- Molecular Biology
- Molecular and Cell Biology
- Cell Biology
- Biochemistry
- Genetics and Genomics
- Ecology
- Biology-Other

C Evaluation Details

C.1 Prompt

C.2 Sampling Parameters

As depicted in Table 6, we report the sampling parameters adopted in the evaluation procedure.

C.3 Evaluation Cost Estimation

We estimate the evaluation cost using the following equation:

$$C_{\text{eval}} = \alpha \left(c_{\text{in}} \cdot T_{\text{in}} + c_{\text{out}} \cdot T_{\text{out}} + c_{\text{tool}} \cdot N_{\text{tool}} + c_{\text{cache}} \cdot T_{\text{cache}} + c_{\text{infra}} \cdot t_{\text{eval}} \right) \tag{3}$$

where $c_{\text{in}}, T_{\text{in}}$, $c_{\text{out}}, T_{\text{out}}$, and $c_{\text{tool}}, N_{\text{tool}}$ are the unit costs and quantities for input tokens, output tokens, and tool calls (*e.g.*, web search), respectively[5]. $c_{\text{cache}}, T_{\text{cache}}$ denote the cost and volume of cached

[5]Token and tool call pricing are queried via the OpenRouter cost and stats API: https://openrouter.ai/docs/api/reference/overview#querying-cost-and-stats.

Prompt for Rubric-based Evaluation in $OneMillion-Bench

Your response should be in the following format:

Explanation: {your explanation for your answer choice}

Answer: {your chosen answer}

Confidence: {your confidence score between 0% and 100% for your answer.}

Figure 10: System prompt rubric-based evaluation.

tokens (*e.g.*, prompt cache reads and writes), and $c_{\text{infra}}, t_{\text{eval}}$ denote the infrastructure cost rate and total wall-clock evaluation time, capturing compute and storage overhead beyond API billing. The scaling factor $\alpha \geq 1$ provides a buffer against underestimation due to retries, rate-limit handling, and other incidental overhead not captured by the individual billing terms; in practice we set $\alpha = 1.2$.

D Case Examples

D.1 Economics and Finance

Background:
Today is December 20, 2025. Yesterday (December 19), the Bank of Japan (BoJ) announced an interest rate hike during its Monetary Policy Meeting, raising the policy rate from 0.50% to 0.75%. However, market performance was unexpected: the Japanese Yen did not strengthen following the hike; instead, it depreciated significantly from around 155 prior to the meeting, rapidly breaching the 157 mark. Looking back at the entirety of 2025, the Yen exchange rate experienced multiple rounds of volatility. Despite the Bank of Japan releasing hawkish signals multiple times throughout the year and Japan's current account surplus remaining high, the Yen failed to enter a sustained appreciation trend.

Question:
Task Requirements:
Review the phased characteristics of the Yen's depreciation cycle throughout 2025, and attempt to delineate the volatility cycles based on key policy milestones and macroeconomic data.
Conduct a deep analysis of why the Yen exchange rate exhibited a snapback depreciation and fell below 157 after yesterday's (December 19) rate hike to 0.75%—a thirty-year high—and identify the underlying market logic.
Research and integrate changes in Japan's current account structure in 2025 (specifically the primary income surplus and the services trade deficit) to analyze their long-term suppressive effect on the Yen exchange rate.
Provide a strategic outlook for the Yen's trend in 2026 and identify core variables that may trigger a trend reversion or trend reversal for the Yen.

Rubrics

1. Analytical Reasoning
(+8) Divide the 2025 Yen trend into at least 3 distinct phases, with each phase containing: (1) time interval, (2) dominant driving factors, (3) characteristics of exchange rate movement.
(+6) Accurately cite at least 2 of the following core data points: (1) policy rate hike from 0.50% to 0.75%, (2) pre-meeting exchange rate approx. 155, (3) post-meeting rate breaching 157, (4) this represents a thirty-year high.
(+9) Explicitly state that the real interest rate remains negative after the Japanese rate hike, and provide at least one of the following: (1) specific calculation (e.g., 0.75% - 3.0% ≈ -2.25%), (2) comparison with U.S. real interest rates, (3) explanation of why negative real interest rates fail to support the exchange rate.
(+7) Analyze the dovish signals from BoJ Governor Kazuo Ueda's post-meeting remarks, including at least 2 of the following: (1) emphasis that "financial conditions remain accommodative," (2) lack of a clear path for future hikes, (3) market interpretation as "buy the rumor, sell the fact."
(+8) Explain the behavior of Carry Trades post-December 19, including: (1) the low-cost characteristic of the Yen as a funding currency, (2) the logic of arbitrage position re-entry after the hike, (3) mention of strengthening cross pairs (e.g., EUR/JPY, AUD/JPY).
(+8) Point out the structural weaknesses in the composition of Japan's current account surplus, encompass-

ing the following 2 points: (1) primary income surplus is largely reinvested overseas, creating no demand for FX conversion, (2) services trade deficit (especially the digital services deficit) constitutes structural, inelastic Yen-selling flow.
(+8) Provide specific data on Japan's digital services deficit (e.g., approx. 6–6.6 trillion Yen), or detail its composition (cloud services, advertising, software licensing, etc.).
(+8) Analyze the short-term negative shock of capital repatriation, noting: (1) rising JGB yields attract Japanese investors to reduce holdings of overseas assets, (2) this process causes a short-term increase in Yen supply and a demand mismatch, (3) it only supports the Yen in the medium-to-long term.
(-7) Heavy reliance on technical analysis terminology (e.g., Fibonacci retracement, moving average golden cross, RSI oversold) while lacking fundamental logic support.
(+7) When analyzing the different phases of the Yen trend, continuously compare the U.S.–Japan real interest rate spread; the method must include: (1) Japan real rate formula: policy rate - core CPI, (2) U.S. real rate formula: nominal rate (10-year Treasury or Fed Funds Rate) - core PCE/CPI, (3) explicitly state the direction (positive for the U.S., negative for Japan) and magnitude (hundreds of basis points) of the spread.
(+6) List key BoJ meeting dates and resolutions in 2025, such as: Jan 24 (hike to 0.5%), Jun 17 (JGB purchase tapering path), Sep 19 (ETF disposition), Oct 30 (split vote), Dec 19 (hike to 0.75%).
(+6) Cite Shunto wage growth data for 2024 and 2025 are $> 5\%$, and conclude based on this that while 2026 wage growth will remain historically high, it will fall below 5% due to the marginal slowdown in inflation.
(+5) Analyze how Japan's aging population affects long-term capital flow patterns, noting: aging → pension funds liquidating overseas assets → structural shift from "capital exporting nation" to "capital repatriating nation."
(-4) Heavy use of second-hand phrasing such as "reportedly," "market rumors," or "analysts expect," without direct citation of official BoJ documents, Ministry of Finance statements, or primary economic data reports.
(-5) Failure to analyze the deeper impact of the 2 votes supporting a hike (7–2 split) during the October BoJ meeting on market expectations and central bank credibility (e.g., "split vote increases policy uncertainty," "implies rising internal hawkish power").
(-5) Simplistically attributing the post-hike depreciation to "market priced in" or "buy the rumor, sell the fact" without deeply analyzing multiple mechanisms such as real interest rates, policy credibility, and capital flows.

Response

You referenced "today = Dec 20, 2025," but as of my system date Feb 18, 2026, the Dec 19, 2025 BoJ meeting is a past event. Below I treat 2025 as a completed year, and the Dec 19, 2025 hike (0.50% → 0.75%) as the late-cycle shock you described. (investing.com)

1) USD/JPY in 2025: phased volatility / "depreciation-cycle" mapping by milestones

A useful way to segment 2025 is: (A) early-year USD stress → JPY rally, then (B) mid/late-year re-widening of "JPY funding" logic as BoJ normalization stayed slow and Japan's services outflows remained structural.

Phase 1 — Jan: "BoJ hikes, but JPY still a funding currency"

Policy milestone: Jan 24, 2025 BoJ hiked 0.25% → 0.50%. (apnews.com)

Market character: Brief JPY support, but the market quickly reverted to rate-differential + carry framing (JPY still low yield vs. G10).

Phase 2 — Late Mar–Apr: "Tariff/Fed-independence risk → USD de-risking → JPY safe-haven bid"

Macro/policy catalyst: Escalation of US tariff / policy uncertainty and Fed independence concerns triggered "sell USD" hedging.

Price action marker: USD/JPY broke ∼140 (JPY strongest area of 2025). (bloomberg.com)

Interpretation: This was less "Japan strength" and more global USD risk premium rising; JPY benefited as a liquid hedge.

Phase 3 — May–Aug: "Mean reversion into a range; JPY becomes tethered to Fed-cut pricing"

Macro marker: As risk sentiment stabilized intermittently, carry demand returned and USD/JPY drifted back up from the April lows.

Range framing: Research commentary described range trading and high sensitivity to Fed cut expectations rather than Japan's own data. (mufgresearch.com)

Phase 4 — Sep–Oct: "BoJ holds at 0.50% (dissents), market punishes slow normalization"

Policy milestone: BoJ repeatedly held at ∼0.50%, with dissenters favoring hikes.

Price action marker: After one such hold, the yen weakened sharply (example: ∼153 per USD reported). (ft.com)
Interpretation: The market concluded the BoJ reaction function was still highly asymmetric (very gradual hikes, cautious guidance).
Phase 5 — Dec: "Hike delivered, but guidance disappoints → snapback depreciation"
Policy milestone: Dec 19, 2025 BoJ hiked 0.50% → 0.75% (30-year high). (apnews.com)
Price action marker: Yen weakened; USD/JPY spiked through 157 (Reuters cited ∼157.67 intraday). (investing.com)
Important contextual point: Japan's MoF **did not support JPY with actual intervention operations during 2025** (multiple monthly windows show JPY 0). That removed a key "hard downside limiter" for USD/JPY breakouts. (mof.go.jp)
2) Why USD/JPY snapbacked above 157 after the Dec 19 hike: the market logic
The apparent paradox ("hawkish action, weaker currency") is common when the marginal information in the decision is less hawkish than positioning implied.
(1) Priced-in hike + sell the fact
Reuters explicitly noted the move was well-telegraphed, and selling intensified after Ueda stayed vague on timing/pace of future hikes. (investing.com)
Mechanics: pre-meeting investors build JPY-long / USDJPY-short expectations; post-meeting if the statement/presser does not upgrade the expected path, those longs unwind fast.
(2) The path matters more than the level: forward rates did not reprice hawkishly enough
Multiple reports emphasized the BoJ was unclear about the "neutral destination" and still described conditions as accommodative / real rates deeply negative. (barrons.com)
So the market read the hike as one step, not a regime of fast normalization, leaving terminal BoJ rates and speed lower than hoped by yen bulls.
(3) US–Japan rate differentials still favored USD even after Fed cuts
After the Fed's Dec 2025 decision, the fed funds target range was 3.50%–3.75%, far above 0.75% in Japan. (federalreserve.gov)
That spread keeps the carry math attractive unless the market believes the BoJ will keep hiking or the Fed will cut faster.
(4) Cross-yen dynamics + risk-on tone
Reuters reported not only USDJPY strength but also record/major highs in EUR/JPY, CHF/JPY, GBP/JPY on the day—classic broad JPY selling rather than a USD-only story. (investing.com)
This usually aligns with risk assets holding up and investors maintaining/adding JPY-funded exposures.
(5) Fiscal risk channel: higher yields can be "bad JPY" if interpreted as supply/fiscal stress
FT coverage around the post-hike period highlighted concerns about bond yields and future issuance tied to stimulus, i.e., a risk premium problem rather than a growth-positive normalization. (ft.com)
3) Japan's 2025 current-account structure shifts—and why they can suppress JPY long-term
2025 CY headline structure (official BoP)
For calendar year 2025 (preliminary), Japan ran:

- Current account surplus: JPY 31.8799 tn
- Primary income surplus: JPY 41.5903 tn
- Services balance: -JPY 3.3928 tn
- Goods balance: -JPY 0.8487 tn (deficit narrowed a lot vs 2024)

(mof.go.jp)
So you are right that "current account is strong," but the FX impact depends on conversion behavior and offsetting structural outflows.
(A) Primary income surplus ≠ immediate spot JPY buying
The primary income table shows large components that often do not translate into immediate repatriation: big direct investment income, including reinvested earnings (kept abroad by corporates). (mof.go.jp)
FX implication: Even if Japan earns huge overseas income, if it is retained overseas, reinvested, or hedged, it does not create the spot-market yen demand that would force sustained appreciation.
(B) Services deficit widened, and its composition matters
The services deficit deepened in 2025 (-JPY 3.3928 tn). (mof.go.jp)
Inside services, 2025 shows: travel surplus +JPY 6.3429 tn (strong inbound tourism support), but large structural deficits in "modern" services, notably telecom/computer/info services (-JPY 2.4740 tn) and other business services (-JPY 5.9443 tn), plus insurance & pension services (-JPY 3.3137 tn). (mof.go.jp)
Why this can suppress JPY over time: These service categories are recurring payments (subscriptions, cloud, advertising, consulting, reinsurance, etc.). They are less sensitive to short-term FX moves and can behave like a structural demand for foreign currency, diluting the "surplus ⇒ stronger currency" intuition.
4) Strategic outlook for 2026: what could trigger reversal vs reversion?
Where the market started 2026

USD/JPY in early 2026 traded roughly 152–159, and was around the low-153 area in mid-Feb 2026 (per published daily history). (poundsterlinglive.com)
Base case (range-to-lower USD/JPY; gradual JPY recovery)
A reasonable base case is JPY modestly stronger over 2026 if: (1) BoJ keeps hiking (or credibly signals a path), not just one-off moves, (2) the Fed continues easing (narrowing the US–Japan spread), and (3) global risk sentiment is not persistently risk-on carry.
Recent reporting suggested BoJ signaled additional hikes likely in 2026 even when holding 0.75%. (ft.com)
What would trigger a trend reversion (JPY strengthening sustainably)
Core variables to watch:

- BoJ reaction function turns symmetric: clear guidance on pace and a believable estimate of "neutral" (reduces the chance of another "hike but dovish" selloff). (barrons.com)
- Japan wage/inflation persistence (Shunto + services inflation): if wage growth sustains and BoJ can hike without choking demand, JPY can re-rate higher.
- Fed cuts faster than currently discounted: a sharper US slowdown compressing the front-end spread is the cleanest "JPY bull" macro setup. (federalreserve.gov)
- Risk-off / deleveraging shock: any global volatility event that forces carry unwind tends to produce abrupt JPY strength (as in April 2025's USD de-risking episode). (bloomberg.com)
- Meaningful improvement in the services deficit: e.g., sustained shrinkage in digital/business-services outflows (hard, but it is the structural piece).

What would trigger a trend reversal against JPY (USD/JPY back toward highs)

- BoJ slows/pauses again due to growth or political pressure (market re-anchors on "JPY funding").
- Japan fiscal stress narrative rises (issuance fears → higher yields but wider risk premium) rather than "healthy normalization." (ft.com)
- Persistent risk-on environment (carry demand stays dominant).
- No intervention deterrent even during sharp moves (2025 had JPY 0 operations in multiple windows). (mof.go.jp)

If you want, I can turn the 2025 phase section into a timeline table (date → event → USD/JPY reaction → which flow dominated: rates / risk / BoP / positioning) and tailor it to the exact USD/JPY series you use (spot close, Tokyo close, or intraday highs/lows).

D.2 Law

Background:
Case Background:
A company incorporated in the United States (hereinafter referred to as the "Plaintiff") and a company incorporated in Singapore (hereinafter referred to as the "Defendant") entered into a commodity purchase and sale contract in 2024. The Plaintiff acted as the Supplier and the Defendant as the Purchaser. After the Plaintiff fulfilled its delivery obligations, the Defendant failed to make payment as agreed. Despite multiple collection attempts by the Plaintiff, the Defendant refused to pay. Consequently, the Plaintiff filed a lawsuit against the Defendant in the Singapore courts.
Upon receiving the notice of action, the Defendant, having failed to retain a copy of the contract, requested a copy of the contract for review from the Plaintiff. After review, the Defendant filed an application with the court to stay the proceedings on the following grounds: Article 20 of the contract provides that "This Contract shall be governed by, and construed and enforced in accordance with the laws of the State of Ohio. The parties hereby agree to the courts of Ohio as the chosen forum and submit to their jurisdiction. Under no circumstances shall the Supplier be subjected to the jurisdiction of foreign courts without its prior written consent." The Defendant argues that the Plaintiff, as the Supplier, did not obtain its own prior written consent before initiating the lawsuit in the Singapore court, which constitutes a violation of Article 20 of the contract; therefore, jurisdiction over this case should lie with the courts of Ohio, USA.
The Plaintiff rejects the Defendant's opinion and insists on proceeding with the litigation in Singapore.
The Defendant further invokes the doctrine of forum non conveniens, arguing that since the contract chooses the application of the laws of the State of Ohio, it is evident that the Singapore court is a forum non conveniens and should decide not to hear this case.
Question:
Acting as the Presiding Judge of the Singapore court in this case, you are required to adjudicate whether this case should continue to be heard in Singapore. Please focus your argumentation on the following issues:

1. Is the Plaintiff required to obtain its own written consent?
2. Does the doctrine of forum non conveniens preclude the Singapore court from exercising jurisdiction?
3. Are there any other facts and reasons to support your analysis?

Rubrics

1. Factual Information

(+2) Correctly distinguished the procedural status of both parties in the response, specifically identifying the Plaintiff as a US company and the Defendant as a Singaporean company.

2. Analytical Reasoning

(+3) Explicitly provided a general conclusion of "dismissing the Defendant's application for a stay of proceedings" or "continuing the trial in Singapore" at the beginning or end of the response.
(+5) Comprehensively enumerated the three core components of Article 20 of the contract (governing law, choice of forum, protection of the Supplier), concluding that Article 20 constitutes a permissive jurisdiction clause rather than an exclusive jurisdiction clause.
(+3) Determined that the literal meaning and purpose of Article 20 are to protect the unilateral interest of the Supplier (Plaintiff) in choosing the forum, i.e., determining the clause as an exclusive right conferred upon the Supplier.
(+3) Pointed out that the Plaintiff's act of filing a lawsuit in Singapore itself constitutes a waiver of the right requiring "its own written consent".
(+7) Clarified the definition of the doctrine of forum non conveniens, comprising core elements: the existence of a competent court, that trying the case would cause inconvenience to the parties and the judicial system (or fail to ensure the fair administration of justice), and the existence of a more appropriate alternative forum.
(-6) In analyzing the applicability of the Hague Convention on Choice of Court Agreements or choice of court agreements, erroneously stated the contracting states, specifically by claiming that Singapore is not a party to the Hague Convention on Choice of Court Agreements, or that the United States (Ohio) is a party thereto.
(-3) Failed to distinguish the independence of governing law from venue/forum, supporting the forum non conveniens principle solely on the grounds of "applying Ohio law".
(+7) Elucidated that both Singapore and Ohio belong to the common law system; therefore, applying Ohio law presents no substantive difficulty, and the parties have not argued that the application of different legal principles by Ohio courts versus Singapore courts would substantively affect the dispute.
(+3) Analyzed that adjudicating in Singapore is more conducive to the enforcement of the judgment, given that the Defendant's assets are likely located in Singapore.
(+3) Pointed out that the mere fact that the lex causae (governing law) is a foreign law (Ohio law) is insufficient to constitute grounds for applying the doctrine of forum non conveniens.
(+5) Noted the fact that the Defendant did not retain a copy of the contract to infer that Article 20 was not subject to close negotiation between the parties, thereby reducing the influence or weight of said clause.
(+4) Clarified that the Singapore court has personal jurisdiction (in personam jurisdiction) over the Defendant.
(-2) Erroneously allocated the burden of proof regarding the doctrine of forum non conveniens, arguing that the Plaintiff should prove that Singapore is the appropriate forum.

3. Instructions Following

(+3) Adopted the identity and tone of a Presiding Judge of the Singapore Court for the judicial reasoning; the narrative perspective should be that of the judge (using "I", "the Court", or avoiding self-reference without using a researcher or third-party tone).
(-2) When discussing "other facts and reasons for argument," completely repeated the arguments used in answering the two questions: "whether the Plaintiff must obtain its own written consent" and "whether the forum non conveniens doctrine hinders jurisdiction" in the previous sections.

4. Structure and Formatting

(+2) The structure of the response must at least contain "General Conclusion, Issue 1, Issue 2, Issue 3", where Issues 1, 2, and 3 refer to the analysis of whether the Plaintiff must obtain its own written consent, whether the doctrine of forum non conveniens impedes Singaporean jurisdiction, and whether there are other supporting facts and reasons.

Gemini-3.0-pro Response

JUDGMENT
Case: Plaintiff (US Supplier) v. Defendant (Singapore Purchaser)
Forum: High Court of the Republic of Singapore
Presiding Judge: [Assigned Judge]
DECISION
The Defendant's application to stay the proceedings is **dismissed**. This Court retains jurisdiction, and the case shall proceed in Singapore. The reasons for this decision, addressing the specific issues raised regarding the interpretation of the jurisdiction clause and the doctrine of forum non conveniens, are set out below.
General Conclusion
The Defendant's application is dismissed. The action will continue in Singapore.
Issue 1. Is the Plaintiff required to obtain its own written consent?
No. The Defendant's interpretation of Article 20 is commercially absurd and legally untenable.
The Defendant argues that the Plaintiff (the Supplier) breached Article 20 by failing to obtain its own "prior written consent" before filing suit in Singapore. This Court rejects this submission for the following reasons:

- **Nature of the clause (benefit of the party):** The sentence in Article 20 stating, "Under no circumstances shall the Supplier be subjected to the jurisdiction of foreign courts without its prior written consent," is an asymmetrical jurisdiction clause designed exclusively for the benefit of the Plaintiff (the Supplier). Its purpose is to protect the Supplier from being dragged into a foreign jurisdiction against its will by the Purchaser. It acts as a shield for the Supplier, not a shackle.
- **Waiver by conduct:** A party for whose benefit a contractual provision is inserted is entitled to waive that benefit. By voluntarily commencing proceedings in the Singapore courts, the Plaintiff has unequivocally waived its right to object to this forum.
- **Absurdity of "self-consent":** It is a logical fallacy to suggest that a legal entity must issue a formal written document to itself granting permission to take an action it has voluntarily decided to undertake. The act of filing the Writ of Summons and Statement of Claim constitutes the clearest possible evidence of the Plaintiff's consent to submit to the jurisdiction of the Singapore courts. To require a separate internal written memo would be an exercise in futility and form over substance.

Therefore, the Plaintiff was not required to generate a prior written consent to itself to validate the proceedings.
Issue 2. Does the doctrine of forum non conveniens preclude the Singapore court from exercising jurisdiction?
No. The Singapore Court is the appropriate forum for this dispute.
The Defendant invokes the doctrine of forum non conveniens (based on the principles in *Spiliada Maritime Corp v Cansulex Ltd*, applicable in Singapore), arguing that the choice of Ohio law makes Singapore an inappropriate forum. This argument fails for the following reasons:

- **Distinction between choice of law and choice of forum:** The fact that the contract is governed by the laws of the State of Ohio does not automatically render the courts of Ohio the only appropriate forum. It is a common occurrence for Singapore courts to adjudicate disputes governed by foreign laws. The choice of law is a factor to be considered, but it is not decisive.
- **Nature of the dispute:** This is a straightforward debt collection case arising from a commodity sale where delivery has occurred, and payment is outstanding. The legal issues likely revolve around contractual performance and payment obligations, which are not so uniquely complex under Ohio law that a Singapore court cannot adjudicate them (potentially with the aid of expert evidence on Ohio law, if strictly necessary).
- **Connecting factors:** The Defendant is a company incorporated in Singapore. As such, Singapore is the Defendant's home jurisdiction. The Defendant is clearly present here, and the Singapore courts have personal jurisdiction over the Defendant. Therefore, Singapore is a prima facie natural forum for the dispute.

Consequently, the Defendant has failed to demonstrate that Ohio is a "clearly more appropriate" forum than Singapore for the interests of all parties and the ends of justice.
Issue 3. Other facts and reasons supporting this analysis
In addition to the points above, the Court relies on the following factors to deny the stay:

- **Enforceability and efficiency:** The ultimate goal of litigation is to obtain an enforceable judgment. Since the Defendant is a Singapore-incorporated entity, its assets are presumably located in Singapore. If this Court were to stay proceedings in favor of Ohio, the Plaintiff would have to litigate in the US, obtain a judgment there, and then return to Singapore to register and enforce that foreign judgment. This would result in unnecessary circularity, delay, and expense. Hearing the case in Singapore allows for a direct judgment and immediate enforcement, serving the ends of justice and judicial economy.

- **Burden of proof:** In a case where the Plaintiff sues the Defendant in the Defendant's own jurisdiction (Singapore), the burden on the Defendant to show that the forum is inappropriate is particularly high. The Defendant has offered no evidence regarding the location of witnesses or evidence that would make a trial in Ohio significantly more convenient.
- **Interpretation of "courts of Ohio":** While Article 20 mentions the parties agree to Ohio courts, it does not use the word "exclusive" regarding the Plaintiff's obligations. When combined with the "consent" proviso, the clause is best interpreted as: (a) exclusive jurisdiction for suits against the Supplier (unless it consents otherwise), and (b) non-exclusive jurisdiction for suits by the Supplier against the Purchaser. The Plaintiff has exercised its option to sue in the Defendant's domicile.

Conclusion
The application for a stay of proceedings is dismissed with costs awarded to the Plaintiff. The Defendant is ordered to file its Defence within the statutory timeline.

D.3 Healthcare and Medicine

Background:
You are an orthodontist. You have received a 10-year-old female patient whose parents' chief complaint is dental misalignment and "protrusive maxillary incisors" (buck teeth), which have gradually worsened over the past 2 years. There is no family history of similar facial features.
Upon taking the medical history, it was found that the patient has a habit of nocturnal mouth breathing. A visit to the Otolaryngology department revealed adenoid hypertrophy and severe Obstructive Sleep Apnea (OSA); drug therapy has shown no significant improvement after 3 months.
Extraoral examination reveals a convex profile, maxillary protrusion, mandibular retrusion, and a long lower facial third. Intraoral examination reveals: mixed dentition, moderate crowding of upper and lower anterior teeth, maxillary arch constriction, Grade II increased overjet, Grade I deep overbite, and bilateral Angle's Class II molar relationship.
Cephalometric analysis shows a distinct concavity on the lower border of the C2 vertebral body, while the lower borders of C3 and C4 are flat. Measurements indicate SNA$= 80^\circ$, SNB$= 74^\circ$, SN–MP$= 32^\circ$, U1–PP$= 120^\circ$, and L1–MP$= 90^\circ$.
After researching online and consulting other clinics, the family has presented the following demands:

- Non-extraction treatment only; they claim extraction will cause teeth to become spaced, mobile, and prone to loss later in life, affecting eating;
- No adenoidectomy, as they believe it will affect immunity;
- Request to wear silicone-based appliances, strictly for night use only, so as not to affect daytime study.

Question:
Please provide professional treatment recommendations based on the above information and incorporating the latest clinical guidelines or expert consensus from 2024–2025.

Rubrics

1. Factual Information
(+5) Diagnose Angle Class II malocclusion based on the distal molar relationship.
(+5) Diagnose skeletal Class II malocclusion based on SNA, SNB, and ANB values.
(+7) Determine that development is at the CS2 stage based on cervical vertebral morphology, indicating the patient is prior to the peak growth spurt.
(+5) Recommend adenoidectomy based on severe Obstructive Sleep Apnea (OSA), inefficacy of drug therapy, and the presence of mouth breathing habits.
(+5) Cite the "Expert Consensus on the Clinical Application of Silicone Oral Functional Trainers (2024)" to recommend against the use of silicone-based appliances.

2. Analytical Reasoning
(+5) Analyze and identify mouth breathing caused by adenoid hypertrophy as the primary etiology of the patient's dentofacial deformity.
(+6) Propose maxillary expansion therapy to resolve the maxillary arch constriction.
(+7) Recommend using a mandibular advancement appliance to guide mandibular growth, with Twin Block appliance being the preferred choice.

(+5) Recommend full-day wear for the functional appliance, refuting the parents' request for night-only wear, and explain the principles of muscular, skeletal, and dental adaptation.
(+5) Propose myofunctional training (e.g., tongue muscle, orbicularis oris muscle training).
(+5) Explain the safety of extraction treatment, clarifying that spaces will close and properly executed orthodontic treatment will not cause pathologic tooth mobility or tooth loss.
(-5) During expansion therapy, failing to combine a Twin Block appliance with a palatal expander (e.g., jackscrew) for simultaneous expansion and mandibular advancement, and limiting treatment to a single expansion method (rapid/slow) or a single type of expander.
(-8) Yielding to the family's unreasonable demands by considering the use of silicone-based appliances for treatment.

3. Instructions Following
(+7) Address the parents' refusal of surgery by pointing out that orthodontic treatment is only suitable after the mouth breathing habit is resolved.
(+3) Address the parents' concerns about extraction by explaining the evolutionary trend leading to insufficient bone mass relative to tooth mass.
(+5) Explicitly recommend against silicone-based appliances, highlighting disadvantages such as insufficient hardness, poor retention, and lack of skeletal remodeling effects.

4. Structure and Formatting
(+8) Structure the orthodontic treatment plan into two stages: Phase I functional treatment and Phase II finishing and detailing.
(-5) The response contains 3 or more errors in the use or translation of professional terminology.
(-5) The response uses large, dense blocks of pure text lacking clear subheadings or bullet points, resulting in poor readability.

5. Others
(-7) Omitting the disclaimer that AI only provides suggestions and offline medical consultation is required.

GPT-5.2-High-Search Response

1) Problem list & priorities (what matters most first)
This 10-year-old has (a) severe pediatric OSA with adenoid hypertrophy and chronic mouth breathing, plus (b) skeletal/dental Class II div 1 (mandibular retrusion with proclined maxillary incisors, increased overjet), (c) transverse maxillary constriction, (d) moderate anterior crowding, and a vertical tendency/long lower facial third.
From a medical and risk-management standpoint, severe OSA is the first priority because untreated pediatric OSA is linked to neurocognitive/behavioral, cardiovascular and quality-of-life consequences; orthodontics should be coordinated around airway care, not used as a substitute.
Your cervical vertebrae maturation finding (concavity only at C2; C3–C4 flat) is consistent with early CVM (pre-peak growth), meaning some growth modification can start now, but the most efficient mandibular "growth spurt" window is usually a bit later (often when C3 also shows concavity). This supports a staged (Phase I / Phase II) plan.
2) Airway / OSA recommendations (based on 2024–2025 guidance)
A. Re-counsel: surgery is standard first-line for hypertrophy-driven pediatric OSA
Major pediatric guidance continues to recommend adenotonsillar surgery as first-line when hypertrophy is present, and CPAP when surgery is not performed or is ineffective. (publications.aap.org)
For severe OSA, ENT guidelines also emphasize careful peri-operative planning/monitoring and that OSA can persist and needs follow-up. (aafp.org)
Clinical recommendation: I would formally recommend re-discussion with ENT + pediatric sleep medicine about adenoidectomy (often with tonsil evaluation) as the evidence-based first-line option, with a plan for post-treatment reassessment (repeat PSG or objective testing).
B. If the family refuses adenoidectomy: CPAP is the medically appropriate fallback
If adenoidectomy is declined, then CPAP is the most evidence-based non-surgical therapy for significant OSA (though adherence can be difficult). This "CPAP if surgery not performed" pathway is explicitly recommended in pediatric guidance. (publications.aap.org)
Clinical recommendation: If no surgery, I would refer for CPAP initiation/titration and adherence coaching before relying on orthodontic appliances for an airway problem.

C. Orthodontics can be an adjunct for selected airway anatomy (maxillary constriction)
The 2024 American Thoracic Society (ATS) guideline on persistent pediatric OSA states that children with persistent OSA and specific craniofacial features may be candidates for orthodontic/dentofacial orthopedic treatment, and the panel specifically notes that RME is preferred when there is a constricted maxilla (high/narrow palate, often posterior crossbite). (pmc.ncbi.nlm.nih.gov)
Also, a 2025 randomized controlled trial comparing RME vs adenotonsillectomy found both improved PSG parameters (including AHI), while adenotonsillectomy produced better symptom/QOL improvement. (pubmed.ncbi.nlm.nih.gov)
So: RME is reasonable for this child's narrow maxilla and may help airway metrics—but it is not a guarantee, and for severe OSA it should be adjunctive, not the sole treatment.
3) Orthodontic plan (what I would recommend clinically)
Phase I (now; mixed dentition, narrow maxilla, trauma-risk overjet)
1. Rapid Maxillary Expansion (RME) with a fixed expander (Hyrax/Haas-type).
Goals: correct constriction, gain arch width/perimeter to help crowding, improve nasal airflow potential, and create a better foundation for Class II correction.
Important: RME is not "night-only." It must be worn 24/7 because it is cemented and works by orthopedic forces over days–weeks, then retained to stabilize.
2. Incisor/overjet risk reduction
Because protrusive incisors raise the risk of trauma, I would usually add either:

- a short "2 $\times$ 4" partial fixed appliance to align/level incisors and manage deep bite, and/or
- a functional Class II approach once transverse issues are controlled (see below).

3. Class II correction (mandibular retrusion) – realistic options

- Most predictable: a fixed functional appliance (e.g., Herbst-type) later in Phase I/early Phase II—minimal daytime compliance burden.
- Removable functional (Twin-Block-type): workable if worn enough hours.

Evidence on wear time is mixed; an RCT found part-time wear ($\sim$9 h/day) could achieve similar overjet reduction to full-time in 12 months (pubmed.ncbi.nlm.nih.gov), but other work using compliance indicators suggests greater skeletal effect with longer daily wear (pubmed.ncbi.nlm.nih.gov).
Bottom line: "night-only" may improve overjet some, but it is less reliable for meaningful skeletal correction and stability.
Phase II (permanent dentition, around/after growth peak)
Comprehensive braces/aligners as appropriate to:

- finalize Class II correction (as needed),
- fully relieve crowding and coordinate arches,
- optimize incisor inclination and overbite.

4) Addressing the family's three demands (with professional boundaries)
Demand 1: "Non-extraction only; extractions cause spacing, mobility, tooth loss later."
Professional response:

- Extractions do not inherently cause teeth to become loose and fall out later. Long-term tooth loss risk is driven primarily by caries, periodontal disease, and maintenance, not by having had orthodontic extractions. Temporary mobility during orthodontic movement can occur in any orthodontic treatment (extraction or non-extraction) and usually resolves with retention and healthy periodontium.
- Spaces after extraction are a treatment-execution/retention issue, not an inevitable biologic consequence. If spaces are properly closed with correct root positioning and then retained, they are usually stable; if spaces reopen, it is managed orthodontically.

Demand 2: "No adenoidectomy; it will affect immunity."
Professional response:

- There is substantial evidence (including systematic review/meta-analysis and broad reviews) indicating no clinically significant long-term impairment of humoral/cellular immunity after tonsil/adenoid surgery in typical children, and that remaining lymphoid tissue can compensate. (pubmed.ncbi.nlm.nih.gov)
- At the same time, there are large observational data suggesting associations between removal and higher long-term rates of some respiratory/allergic/infectious diagnoses—this does not prove causation for an individual child, but it is part of balanced counseling. (jamanetwork.com)
- For a child with severe OSA, the known morbidity of leaving OSA untreated is typically considered to outweigh the uncertain/variable long-term infection-risk associations—this is why adenotonsillar surgery remains standard first-line when hypertrophy is present. (publications.aap.org)

Demand 3: "Silicone appliance, night-only, not affecting daytime study."
They are likely referring to prefabricated silicone myofunctional appliances (e.g., trainer-type).
Non-negotiables (professional boundary)

- I cannot ethically recommend night-only silicone orthodontics as the sole management for severe pediatric OSA.
- If adenoidectomy remains refused, I would require documentation that the family understands that CPAP is the medically recommended non-surgical treatment, and that orthodontics is adjunctive. (publications.aap.org)

D.4 Natural science

Background:
A Quantum Spin Liquid (QSL) is a state of matter characterized by the absence of long-range magnetic order and the presence of high degrees of quantum entanglement. Experimental research on Herbertsmithite has revealed that its spin correlation behavior transcends the theoretical framework of simple nearest-neighbor models. Although the ground state of Herbertsmithite does not exhibit traditional long-range magnetic ordering at low temperatures and displays behavior similar to short-range Resonating Valence Bond (RVB) states, no spin gap is observed in Q–ω space. This exotic phenomenon necessitates the consideration that the spin liquid state in Herbertsmithite may possess more complex spin correlation patterns.
Question:
1. Referencing results from relevant Inelastic Neutron Scattering (INS) experiments, elucidate the basis for determining that spin correlation behavior in Herbertsmithite exceeds the framework of the nearest-neighbor model.
2. Analyze the impact of these spin correlations, which transcend nearest-neighbor interactions, on the properties of the Herbertsmithite ground state spin liquid.
3. Combining relevant INS data, discuss why the spin correlation pattern in Herbertsmithite does not conform to the simple short-range Resonating Valence Bond (RVB) model, and propose the challenges this phenomenon poses to theoretical models.

Rubrics

1. Analytical Reasoning
(+5) In INS experiments on Herbertsmithite, the observed spin excitation modes exhibit significant discrepancies with the nearest-neighbor singlet model. For example, the experimental energy-integrated structure factor indicates that the spin correlation behavior in Herbertsmithite demonstrates longer-range correlations.
(+10) Model deviation of the equal-time structure factor: INS experiments indicate that spin correlations are not confined to nearest neighbors but exhibit longer-range spatial correlations, leading to a reduction in the diffuse nature of scattering peaks.
(+5) Supplementary evidence: Although the experimentally measured equal-time structure factor integrated over energies from 1 to 9 meV shows similarity to calculations based on the uncorrelated nearest-neighbor singlet model, the width of the experimental scattering signal in reciprocal space is significantly narrower.
(+4) Mention that at specific momentum positions (e.g., (0,2,0) as described in doi: 10.1038/nature11659), the nearest-neighbor singlet model fails to explain the signal contribution at this location, illustrating the existence of non-nearest-neighbor interactions.
(+6) Provide evidence: Using the momentum position labeling convention from doi: 10.1038/nature11659 as a reference, line scans along the Brillouin zone (0,K,0) direction (1–7 meV) show distinct scattering intensity at positions such as (0,2,0).
(+4) Explain that in the low-energy interval of 0.25–1 meV, extra broad peaks appear at reciprocal space (1,0,0) (using the momentum position labeling convention from doi: 10.1038/nature11659) and equivalent positions; these are not Bragg peaks and cannot be explained by nearest-neighbor interactions.
(-20) The model mistakenly attributes the broad peaks appearing in the low-energy region to interlayer impurity Cu^{2+}.
(+5) Explain the energy insensitivity of scattering intensity: A similar hexagonal ring scattering pattern is observed across the 1.5–11 meV range in INS experiments.
(+2) INS experimental data for Herbertsmithite show almost no change in scattering characteristics over a wide energy range, manifesting characteristics of an energy-insensitive continuum. It must be emphasized that this contrasts with traditional magnets dominated by nearest-neighbor interactions, which typically exhibit marked variations in scattering signatures with changing energy (i.e., they are energy-sensitive).
(+2) Mention one of the critical features of quantum spin liquids: the absence of static order and spin freezing in the ground state.

(+2) Mention one of the critical features of spin liquids: the presence of fractionalized excitation characteristics.
(+6) Stability of the absence of long-range magnetic order: The geometric frustration of the kagome lattice inherently hinders the formation of long-range magnetic order, and spin correlations beyond nearest neighbors further suppress the tendency toward local magnetic moment ordering.
(-10) The model incorrectly assumes that on a kagome lattice, spin correlations beyond nearest neighbors would further promote rather than further suppress the ordering of local magnetic moments.
(+3) Supplementary evidence: In INS experiments, no magnetic ordering was observed even when the temperature was lowered to 0.05 K, and the INS signal lacked sharp spin wave peaks, indicating that long-range correlations maintain the disordered ground state of the spin liquid through quantum fluctuations.
(+7) Support for fractionalized excitation continuum features: INS experiments observe a broad spin excitation continuum band in the 2–11 meV range, rather than the sharp dispersion surfaces of traditional magnets; this is a hallmark of spinon fractionalized excitations.
(+5) Maintaining long-range quantum coherence: The model should point out that spin correlations beyond nearest neighbors contribute to maintaining the long-range quantum coherence of the ground state, and one of the core features of spin liquids is specifically "short-range correlations but long-range coherence."
(+7) Ubiquity of gapless excitations: Short-range RVB corresponds to a gapped spin liquid, whereas experiments observed no spin gap across the entire measured momentum range at energies as low as 0.25 meV.
(-5) The model erroneously analyzes the experimental phenomenon where "scattering intensity rises significantly as energy decreases below 1.5 meV, and remains flat in the 1.5–11 meV range," mistaking this as confirmation of the short-range RVB gap prediction.
(+7) Need to explain that the short-range RVB model (based on random arrangements of local singlets) cannot explain the narrowing of INS peak widths and signals at specific momenta; instead, longer-range correlations are required to modulate scattering.
(-10) Regarding the narrowing of scattering peak widths and the scattering signal at the (0,2,0) position in experiments, the model erroneously uses short-range assumptions to explain the experimental phenomena, ignoring that these phenomena actually indicate the existence of longer-range spin correlations or more complex interactions.
(+7) Requirement for correction of the kagome lattice Hamiltonian: Experiments indicate that non-nearest-neighbor interaction terms must be introduced on top of the nearest-neighbor Heisenberg model to match scattering data, necessitating the reconstruction of a spin Hamiltonian containing long-range interactions.
(-10) The model mistakenly assumes that scattering experimental data can validate the nearest-neighbor Heisenberg model, whereas in reality, the two do not match.
(+6) Controversy over the excitation mechanism of gapless spin liquids: Experiments observe gapless characteristics across the full momentum range, requiring the development of new theories to explain how long-range correlations in two-dimensional systems modulate the dispersion behavior of spinons to form a global gapless continuum.
(-20) The model mistakenly attributes the gapless characteristics observed across the full momentum range in experiments to confirmation of traditional theories.
(+8) Unclear coupling mechanism between correlation length and quantum coherence: Experiments capture signs of the coexistence of "short-range spin correlation + long-range quantum coherence," but existing theories struggle to quantify the regulation laws of non-nearest-neighbor correlations on quantum coherence length.
(-10) The model incorrectly assumes that the theoretical mechanism coupling correlation length and quantum coherence is clear, ignoring the importance of establishing quantitative relationships between correlation range, entanglement entropy, and excitation spectra.

Claude-Opus-4.6 Response

Spin Correlations in Herbertsmithite: Beyond Nearest-Neighbor Models

1. INS evidence that spin correlations exceed the nearest-neighbor framework

Inelastic Neutron Scattering (INS) experiments on single crystals of Herbertsmithite ($ZnCu_3(OH)_6Cl_2$), including the widely cited study in doi: 10.1038/nature11659, provide direct evidence that the spin correlations cannot be captured by a simple nearest-neighbor singlet or nearest-neighbor Heisenberg picture. The key points are:

(a) Equal-time / energy-integrated structure factor deviates from nearest-neighbor expectations.
In a nearest-neighbor singlet (uncorrelated dimer) model, the equal-time structure factor is expected to be broadly diffuse in reciprocal space, with a width set by correlations limited to one lattice spacing.

Experimentally, the energy-integrated structure factor shows a similar overall pattern but with **significantly narrower** features in reciprocal space, implying **longer-range spatial correlations** than nearest neighbors alone.

(b) Specific momentum-space features inconsistent with nearest-neighbor singlets.

Using the momentum labeling convention in doi: 10.1038/nature11659, line scans along the Brillouin zone direction (0,K,0) integrated over 1–7 meV reveal distinct scattering intensity at positions such as (0,2,0). The nearest-neighbor singlet model does not produce a signal contribution at (0,2,0), indicating that correlations beyond nearest neighbors (or additional interaction terms beyond J_1) are required.

(c) Low-energy additional broad peaks not explained by nearest-neighbor interactions.

In the low-energy window 0.25–1 meV, extra broad peaks appear around reciprocal-space positions such as (1,0,0) and symmetry-equivalent points (again using the convention of doi: 10.1038/nature11659). These are not Bragg peaks (no static order) and are inconsistent with a nearest-neighbor-only description of the kagome-plane dynamics, pointing to additional correlations/interactions shaping the low-energy response.

(d) Energy-insensitive continuum behavior across a wide window.

A similar hexagonal ring-like scattering pattern persists across roughly 1.5–11 meV, with the overall scattering characteristics changing very weakly with energy. This "energy-insensitive" continuum contrasts with conventional nearest-neighbor-dominated magnets, where one typically observes energy-sensitive, dispersive spin-wave features and strong energy dependence of the scattering signature.

2. Impact of beyond-nearest-neighbor correlations on the ground state spin liquid

The observed longer-range (non-nearest-neighbor) correlations modify both the stability and the excitation content of the ground state:

(a) Reinforcing the absence of static order and spin freezing.

The kagome lattice is geometrically frustrated, intrinsically suppressing long-range magnetic order. Importantly, correlations beyond nearest neighbors in this setting tend to **further suppress** local-moment ordering tendencies by enhancing competing quantum fluctuations and frustration pathways, helping stabilize a disordered QSL ground state. Consistent with this, INS observes no sharp spin-wave modes and no magnetic ordering even down to very low temperatures (reported as low as $T \approx 0.05$ K in the INS literature), supporting a ground state maintained by strong quantum fluctuations rather than conventional order.

(b) Supporting fractionalized excitations (spinon continuum).

INS detects a broad excitation continuum (not sharp magnon dispersion surfaces) over approximately the 2–11 meV range. Such a continuum is widely interpreted as evidence for fractionalized excitations (spinons) in a QSL, and the detailed momentum modulation of this continuum is sensitive to longer-range correlations and additional exchange terms that reshape the spinon dispersion and spectral-weight distribution.

(c) "Short-range correlations but long-range coherence."

A key qualitative motif for spin liquids is that static correlations can be short-ranged while the quantum state maintains long-range coherence/entanglement. Correlations extending beyond nearest neighbors can contribute to sustaining this coherence by delocalizing singlet/triplet fluctuations over larger loops and distances, thereby stabilizing an entangled ground state without condensing into static order.

3. Why Herbertsmithite does not conform to a simple short-range RVB picture, and challenges to theory

(a) Gaplessness contradicts the simplest short-range RVB expectation.

A simple short-range RVB state built primarily from random nearest-neighbor singlets is typically associated with a **gapped** spin liquid. Experimentally, however, no spin gap is observed across the measured momentum range down to energies as low as $\hbar\omega \approx 0.25$ meV. The persistence of low-energy spectral weight throughout Q–ω space is therefore inconsistent with a basic gapped short-range RVB description.

(b) Peak narrowing and special-momentum intensity require longer-range correlations.

A random nearest-neighbor singlet model would produce a broad, strongly diffuse $S(\mathbf{Q})$ with widths set by one-bond correlations. The experimentally observed narrowing of scattering features in reciprocal space, together with intensity at momenta such as (0,2,0) (doi: 10.1038/nature11659 convention), cannot be explained by purely short-range singlet coverings; instead, longer-range correlations are required to modulate the structure factor and redistribute spectral weight.

(c) Implication: the Hamiltonian must extend beyond the nearest-neighbor kagome Heisenberg model.

To reproduce the detailed $S(\mathbf{Q}, \omega)$, theoretical descriptions generally need to augment the nearest-neighbor Heisenberg model with additional terms such as further-neighbor exchanges (e.g., J_2, diagonal J_d across hexagons) and anisotropies allowed by lattice symmetry (e.g., Dzyaloshinskii–Moriya interactions). The INS mismatches therefore motivate reconstructing an effective kagome-plane Hamiltonian containing non-nearest-neighbor interactions.

(d) Challenges posed to theory.

The experimental combination of (i) a globally gapless continuum, (ii) weak energy dependence of the scattering pattern over a broad window, and (iii) momentum-space features implying longer-range correlations, raises several open issues:

- **Gapless spin-liquid mechanism:** How do longer-range correlations in a 2D kagome system shape the spinon dispersion so that a broadly gapless continuum persists across the full momentum range?
- **Quantifying "short-range correlations + long-range coherence":** Experiments suggest coexistence of limited static correlation length with long-range quantum coherence, but a quantitative coupling law between correlation range, entanglement measures (e.g., entanglement entropy), and excitation spectra remains unclear.
- **Model reconstruction and identifiability:** Multiple extended-Hamiltonian candidates can qualitatively match parts of the data; producing falsifiable, high-precision predictions for the full $S(\mathbf{Q}, \omega)$ remains difficult, especially once realistic longer-range terms are included.

D.5 Industry

Background:
You are an expert in machine learning, and your current research focuses on underwater image enhancement. You have designed an encoder–decoder architecture and trained the model using Mean Squared Error (MSE) as the loss function. During training, you observe that the MSE on the training set steadily decreases, and the MSE on the validation set also decreases accordingly. However, the subjective visual quality on the test set is poor: the enhanced images appear overly smooth and lack fine details. Moreover, the PSNR and SSIM metrics are significantly lower than those of state-of-the-art (SOTA) models.
Question:
Based on the characteristics of the task and the properties of the loss function, analyze the reasons that may have led to this phenomenon.

Rubrics

1. Analytical Reasoning
(+6) Provide an in-depth analysis of the model and clearly point out the fundamental tendency of the MSE loss to drive the network toward predicting the conditional mean of all possible solutions (i.e., the mean solution).
(+4) Explain in detail the concrete effects of this averaging behavior—specifically, how high-frequency details such as edges and textures are transformed into smooth low-frequency components—and explicitly identify this process as the key cause of the over-smoothing phenomenon in the output images.
(+6) Accurately identify the pixel-wise equal-weighting limitation of the MSE loss—namely, that all pixels are assigned identical importance—leading the model to preferentially optimize large-area background regions at the expense of small but critical detail regions.
(+6) Point out that the downsampling operations in the encoder–decoder architecture (such as convolutions with stride and pooling operations) inevitably cause information loss in the image representation.
(+5) Through analysis, clearly state that downsampling operations have an inherent attenuation effect on high-frequency information such as edges and textures.
(+6) Clearly explain the limitations of the decoder's upsampling process, which can only reconstruct images through interpolation based on low-frequency features and is unable to actively recover high-frequency details that have already been lost.
(+6) Explicitly point out that the model may overfit to specific scenes or pixel distributions in the training set, resulting in insufficient generalization performance on previously unseen test scenarios.
(+8) Mention potential issues with the ground-truth labels, including two specific cases: (1) a mismatch between synthetic ground truth and real-world scene characteristics; and (2) misalignment between the labels and the input images.
(-4) After identifying the task as an ill-posed problem, the answer fails to introduce the core mathematical concept for addressing ill-posedness—regularization—and does not discuss how different loss functions provide implicit or explicit regularization.
(-3) The discussion is limited to technical flaws in ground-truth acquisition (e.g., misalignment errors) and fails to reflect on the more fundamental question of "what constitutes ground truth" in underwater image enhancement.
(+4) The answer proposes a three-stage improvement roadmap, such as quick verification (e.g., switching to L1 loss), performance enhancement (e.g., introducing perceptual loss), and the pursuit of state-of-the-art techniques (e.g., GAN-based methods).
(+3) In response to the core attribution that "MSE leads to over-smoothing," the answer designs a simple yet logically rigorous simulation or ablation experiment to demonstrate empirical thinking. An ideal example: "To verify that pixel-wise equal weighting causes background-dominated optimization, the following controlled experiment can be designed: (1) baseline: train with standard MSE."
(+4) Critically point out that not only PSNR and SSIM have limitations, but even current "perceptual

quality” evaluations rely on limited pretrained models (e.g., VGG), and further propose a more systematic, multi-dimensional evaluation framework tailored to underwater image enhancement.
(+3) When proposing improvement strategies (such as using more complex loss functions or architectures), explicitly acknowledge the increased computational cost and discuss feasible trade-off strategies based on application scenarios (e.g., mobile devices or real-time systems).
(-2) The proposed improvement methods are disconnected from the problem attribution and do not explain how the new approaches (e.g., alternative loss functions) mechanistically address MSE-induced over-smoothing and detail loss.
(+4) At the beginning of the answer, the core issue—namely, the MSE loss function—is explicitly identified, without prematurely expanding into excessive background information or parallel arguments.
(+3) Each sub-argument adopts a topic–comment structure, with the first sentence of each paragraph clearly stating the central claim of that paragraph.

2. Structure and Formatting
(-3) The answer merely lists points in a parallel manner, without a clear top-level structure or hierarchical expansion that reflects a coherent main thread (e.g., the progressive relationship among task, loss function, architecture, and data).

Response

The phenomenon you observe stems from the fundamental mismatch between the mathematical properties of Mean Squared Error (MSE) and the perceptual requirements of underwater image enhancement, compounded by the ill-posed nature of the restoration task. Here is a detailed analysis:

1. Regression-to-the-mean effect (primary cause)
Mathematically, minimizing MSE corresponds to estimating the conditional expectation $\hat{y} = \mathbb{E}[y|x]$. Underwater image enhancement is a highly ill-posed inverse problem where multiple plausible clean images (y) can correspond to a single degraded input (x) due to varying water types, turbidity levels, illumination conditions, and intrinsic ambiguities in color recovery. When the model minimizes MSE across this multi-modal distribution, it converges to the pixel-wise average of all possible valid solutions. This averaging process destructively interferes with high-frequency components (textures, edges, fine details), resulting in the observed over-smoothness and “hazy” appearance, as high-frequency details rarely align perfectly across different valid restoration hypotheses.

2. Frequency bias and texture penalty
MSE is dominated by low-frequency components (global color cast, overall brightness) because errors in these channels contribute more to the pixel-wise squared sum. Consequently, the model prioritizes correcting the global blue-green color distortion and mean intensity—where large errors heavily penalize the loss—while suppressing fine details. High-frequency textures (e.g., coral patterns, sediment granules, fish scales) contribute minimally to the total MSE when smoothed, so the network learns to sacrifice these details to minimize variance and noise, generating conservative, blurry predictions that lack the sharp local contrasts essential for realistic underwater scenes.

3. Perceptual-statistical disconnect
MSE assumes pixel-wise independence and treats all errors equally regardless of their structural context. However, the Human Visual System (HVS) is highly sensitive to local contrast, edge coherence, and semantic consistency—properties that MSE does not encode. A model trained with MSE minimizes point-wise deviations without regard for whether the resulting image appears natural or contains coherent textures. This explains why your validation MSE decreases steadily (indicating successful statistical regression) while subjective quality deteriorates: the model achieves the mathematical optimum (minimum variance estimator) that is perceptually suboptimal.

4. Architecture-bottleneck compounding
Your encoder-decoder architecture likely employs downsampling operations (pooling/strided convolutions) that discard high-frequency information during encoding. While skip connections (e.g., U-Net) can mitigate this, MSE-based training provides no incentive for the decoder to reconstruct high-frequency details lost in the bottleneck or to use the skip connections effectively for texture synthesis. Unlike adversarial or perceptual losses that explicitly penalize unrealistic texture statistics, MSE is satisfied by smooth interpolations that “fill in” missing details with local averages, exacerbating the information loss inherent in the architecture.

5. Metric implications (PSNR/SSIM divergence)
Although PSNR is mathematically derived from MSE (PSNR $= 10\log_{10}(\frac{\text{MAX}_I^2}{\text{MSE}})$), your model achieves low PSNR relative to SOTA because:

- **Suboptimal convergence:** The model converges to a high-MSE local minimum characterized by the blurry mean predictions described above. While training MSE decreases, the absolute converged value remains higher than what SOTA methods achieve through better inductive biases or combined loss functions.
- **SSIM specificity:** SSIM measures luminance, contrast, and structure correlations. By optimizing for pixel-wise similarity rather than structural correlation, your model destroys local contrast (variance) and structural coherence—core components of SSIM—resulting in poor scores despite pixel-wise intensity averages being roughly correct.

6. Domain-specific challenges
Underwater images exhibit wavelength-dependent attenuation and backscatter, creating spatially varying degradation where local neighborhoods require non-uniform restoration. MSE enforces a global L_2 penalty that favors spatially uniform solutions, preventing the model from learning the aggressive local contrast enhancements needed to reverse scattering effects, further contributing to the flat, washed-out appearance.
In summary, MSE is intrinsically unsuitable for this task because it seeks the statistically average solution to an ambiguous inverse problem, systematically erasing high-frequency details that define image quality, while providing no mechanism to enforce structural or perceptual plausibility.

Table 6: Sampling parameters adopted in the models evaluation against the $OneMillion-Bench.

Model	Search Provider	Thinking Effort	Misc. Parameters
DeepSeek-V3.2	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
DeepSeek-V3.2-Speciale	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
MiniMax-M2.1	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 65,536
MiniMax-M2.5	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 196,608
Qwen3-235B-A22B-2507	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 262,100
Hunyuan-2.0-Thinking	Tencent	[On by default]	Temperature = 0 Max new tokens = 32,768
Step-3.5-Flash	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 256,000
Doubao-Seed-2.0-Pro-260215	ByteDance	High	Temperature = 0 Max new tokens = 128,000
Doubao-Seed-1.8-251228	ByteDance	High	Temperature = 0 Max new tokens = 64,000
GLM-4.7	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 131,072
GLM-5	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 131,072
Kimi-K2-Thinking	OpenRouter	[On by default]	Temperature = 0 Max new tokens = 262,144
Kimi-K2.5	OpenRouter	High	Temperature = 0 Max new tokens = 262,144
GPT-5.4-High	OpenAI	High	Temperature = 0 Max new tokens = 128,000
GPT-5.2-High	OpenAI	High	Temperature = 0 Max new tokens = 128,000
GPT-5.3-Codex	OpenAI	High	Temperature = 0 Max new tokens = 128,000
Claude-Opus-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Claude-Opus-4.6	Anthropic	High	Temperature = 0 Max new tokens = 128,000
Claude-Sonnet-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Claude-Sonnet-4.5	Anthropic	High	Temperature = 0 Max new tokens = 64,000
Gemini-3-Pro-preview	Google	High	Temperature = 0 Max new tokens = 65,500
Grok-4	xAI	[On by default]	Temperature = 0 Max new tokens = 256,000
Qwen3-Max-2026-01-23	Alibaba	High	Temperature = 0 Max new tokens = 32,768
Qwen3.5-Plus-2026-02-15	Alibaba	High	Temperature = 0 Max new tokens = 65,536

Table 7: Comparison of Expert Score across different domains on the **Global evaluation set**.

Model / Agentic System	Finance		Healthcare		Industry		Law		Natural Science	
	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search
GPT-5.4-HIGH	42.0	68.8 ↑26.8	63.6	53.9 ↓9.7	54.7	58.2 ↑3.5	55.2	55.6 ↑0.5	59.4	59.4 ↓0.0
CLAUDE-OPUS-4.6	42.1	55.5 ↑13.4	68.4	73.7 ↑5.3	58.9	67.5 ↑8.7	53.0	59.5 ↑6.5	52.4	58.9 ↑6.5
GPT-5.3-CODEX	36.2	63.2 ↑27.1	53.2	48.8 ↓4.4	50.4	48.2 ↓2.1	46.7	50.8 ↑4.0	53.8	42.0 ↓11.9
GEMINI-3.1-PRO-PREVIEW	36.3	36.5 ↑0.2	58.2	49.0 ↓9.2	44.3	39.8 ↓4.4	53.6	50.7 ↓2.9	47.3	46.9 ↓0.4
QWEN3.5-PLUS	30.7	31.3 ↑0.6	59.3	60.9 ↑1.6	50.8	52.8 ↑2.0	49.7	48.6 ↓1.1	47.5	51.1 ↑3.6
GPT-5.2-HIGH	29.3	60.7 ↑31.4	59.5	59.3 ↓0.2	50.1	52.5 ↑2.4	48.8	57.0 ↑8.2	50.0	54.7 ↑4.7
KIMI-K2.5	34.6	31.6 ↓3.0	54.8	48.1 ↓6.7	46.0	44.4 ↓1.5	43.8	36.9 ↓6.8	46.4	44.8 ↓1.6
GEMINI-3-PRO-PREVIEW	35.8	47.6 ↑11.8	52.3	61.7 ↑9.3	42.0	50.4 ↑8.4	47.7	50.7 ↑3.0	40.6	53.6 ↑13.0
DEEPSEEK-V3.2-SPECIALE	27.6	28.5 ↑1.0	48.9	45.7 ↓3.1	43.2	37.1 ↓6.1	38.3	36.5 ↓1.8	44.3	45.3 ↑1.0
DOUBAO-SEED-2.0-PRO	25.9	35.8 ↑9.9	51.7	64.3 ↑12.6	38.9	52.6 ↑13.7	35.4	55.0 ↑19.6	45.2	51.3 ↑6.1
GROK-4	27.3	51.4 ↑24.2	45.3	47.5 ↑2.1	39.0	38.5 ↓0.5	39.5	46.4 ↑6.8	42.0	45.6 ↑3.6
GLM-5	30.9	34.5 ↑3.7	47.7	51.9 ↑4.2	36.0	39.1 ↑3.1	38.0	39.6 ↑1.6	39.9	40.5 ↑0.6
MINIMAX-M2.1	19.9	25.4 ↑5.5	46.7	41.4 ↓5.2	39.6	34.6 ↓5.0	31.4	30.8 ↓0.6	36.7	37.3 ↑0.6
HUNYUAN-2.0	25.0	22.6 ↓2.4	46.4	36.7 ↓9.8	34.6	38.3 ↑3.8	28.7	24.9 ↓3.8	38.7	28.4 ↓10.3
STEP-3.5-FLASH	23.5	25.9 ↑2.4	45.3	43.6 ↓1.7	37.9	27.8 ↓10.1	29.2	29.1 ↓0.1	36.0	38.3 ↑2.4
LING-2.5-1T	24.2	22.9 ↓1.3	43.5	40.8 ↓2.7	37.9	36.9 ↓1.0	23.8	25.0 ↑1.1	36.2	30.5 ↓5.7
MINIMAX-M2.5	19.9	16.7 ↓3.3	33.1	38.8 ↑5.7	36.4	39.3 ↑2.9	27.5	26.5 ↓1.0	30.5	36.2 ↑5.7
Deep Research Agents										
O3-DEEPRESEARCH	49.4		50.9		43.9		50.7		36.5	
SONAR-DEEPRESEARCH	45.5		42.7		43.8		41.6		38.0	
O4-MINI-DEEPRESEARCH	40.3		40.3		34.3		36.4		37.1	

Table 8: Comparison of Expert Score across different domains on the **CN evaluation set**.

Model / Agentic System	Finance		Healthcare		Industry		Law		Natural Science	
	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search	Vanilla	Search
GPT-5.4-HIGH	53.6	63.3 ↑9.7	67.3	57.1 ↓10.1	55.8	57.5 ↑1.7	43.9	53.6 ↑9.7	60.6	62.1 ↑1.5
CLAUDE-OPUS-4.6	46.9	60.9 ↑14.0	68.4	70.6 ↑2.2	59.0	69.7 ↑10.7	49.4	60.9 ↑11.4	55.5	60.7 ↑5.2
QWEN3.5-PLUS	48.4	46.9 ↓1.4	60.1	61.7 ↑1.5	50.2	49.0 ↓1.2	48.9	56.1 ↑7.2	49.6	46.7 ↓2.9
GEMINI-3.1-PRO-PREVIEW	47.4	48.6 ↑1.2	59.5	53.9 ↓5.6	43.8	43.3 ↓0.5	45.4	46.2 ↑0.8	48.9	42.7 ↓6.3
GPT-5.2-HIGH	41.4	56.1 ↑14.8	57.0	60.8 ↑3.8	46.4	50.3 ↑4.0	35.0	42.5 ↑7.4	55.8	53.3 ↓2.5
GPT-5.3-CODEX	45.6	54.7 ↑9.1	57.1	54.9 ↓2.2	45.6	54.6 ↑9.0	37.0	41.2 ↑4.3	49.1	49.2 ↑0.1
GEMINI-3-PRO-PREVIEW	46.4	56.4 ↑10.0	53.9	60.3 ↑6.5	42.5	50.9 ↑8.4	40.5	54.7 ↑14.2	46.7	47.6 ↑0.9
KIMI-K2.5	48.8	42.5 ↓6.3	53.0	46.6 ↓6.5	39.2	46.9 ↑7.6	40.3	40.1 ↓0.3	45.4	42.3 ↓3.1
DEEPSEEK-V3.2-SPECIALE	37.1	44.0 ↑6.9	49.9	49.5 ↓0.3	39.6	42.1 ↑2.5	31.0	43.2 ↑12.1	48.1	43.1 ↓5.0
GLM-5	39.5	37.4 ↓2.1	48.5	48.3 ↓0.1	38.3	38.6 ↑0.2	37.1	39.2 ↑2.1	40.2	42.1 ↑1.9
DOUBAO-SEED-2.0-PRO	36.6	48.6 ↑12.0	43.9	54.4 ↑10.6	43.7	50.7 ↑7.0	37.8	51.3 ↑13.5	41.3	47.4 ↑6.1
GROK-4	37.3	43.5 ↑6.2	45.5	45.2 ↓0.3	39.0	36.8 ↓2.2	35.9	30.7 ↓5.2	41.1	41.8 ↑0.6
LING-2.5-1T	29.5	33.6 ↑4.1	46.4	42.8 ↓3.6	40.3	38.9 ↓1.4	31.6	26.3 ↓5.4	35.8	36.4 ↑0.5
HUNYUAN-2.0	36.1	36.6 ↑0.5	37.1	33.2 ↓3.9	41.1	34.1 ↓7.0	35.2	32.1 ↓3.2	32.0	28.3 ↓3.7
STEP-3.5-FLASH	38.5	33.8 ↓4.7	44.2	50.1 ↑5.9	36.1	30.3 ↓5.8	28.4	28.0 ↓0.4	33.8	36.1 ↑2.3
MINIMAX-M2.1	31.6	35.9 ↑4.4	46.2	45.1 ↓1.0	43.3	38.0 ↓5.3	27.6	30.2 ↑2.6	30.2	34.4 ↑4.2
MINIMAX-M2.5	29.4	29.8 ↑0.4	40.3	34.2 ↓6.1	39.4	34.8 ↓4.6	25.0	25.2 ↑0.1	33.0	36.5 ↑3.5
Deep Research Agents										
O3-DEEPRESEARCH	47.4		46.4		40.1		34.9		39.5	
SONAR-DEEPRESEARCH	39.1		42.0		43.1		39.2		37.7	
O4-MINI-DEEPRESEARCH	39.5		39.8		34.7		27.7		38.3	